

# Bounded-Character Fréchet Filter Sites

Filter complexity, Booleanization,  
and three cardinal boundaries

A research report continuing the ProveIt report  
*A two-valued obstruction to gluing Fréchet powers*

Merged from two manuscripts dated 24 September 2026

Let  $\mathbb{P}_\kappa$  be the site of free filters on  $\omega$  generated, modulo the cofinite filter, by fewer than  $\kappa$  sets, and let  $\mathbb{C}_\kappa$  be its dense-site Boolean algebra. Then

$$\begin{aligned} \text{the original completion survives (marked)} &\iff \kappa \leq \mathfrak{p}, \\ \text{no atoms and no geometric points} &\iff \kappa \leq \mathfrak{u}. \end{aligned}$$

For  $\kappa > \mathfrak{p}$  no complete Boolean map fixing the original events exists from the old completion to  $\mathbb{C}_\kappa$  (both sources) or from  $\mathbb{C}_\kappa$  back to it (source 10). The distributivity number  $\mathfrak{h}$  is the exact local joint-decidability boundary of the old completion. The associated reduced-power sheaves are computed by Boolean mixing; they keep finitary transfer, and at the successor stages their ultrafilter specializations are countably saturated.

**Status and scope.** An unrefereed research report, merged from two AI-assisted (ChatGPT) manuscripts, batch 36 manuscripts 08 and 10 of ProveIt's incoming-reports intake. The results are proved in ordinary mathematics in ZFC; they have not been checked in a proof assistant, and placement in a repository that contains Lean and Rocq developments confers no formal status on them. Classical cardinal-invariant and Boolean-valued-model machinery is credited rather than claimed as new. Publication priority has not been established. One relative-consistency application explicitly imports a published forcing model (Millán); its forcing construction is not reproved here.

## Abstract

The ProveIt report on two-valued gluing obstructions proves that the Fréchet reduced-power presheaf is not a sheaf even on the site of countably generated free filters. It computes the associated sheaf on the full filter site as a product of ultrapowers, but that computation cannot be transferred to the smaller site: its ultrafilters are absent. That report explicitly does not assert anything about other filter sites. This report studies them. It merges two manuscripts written independently against the same repository snapshot.

Let  $\mathbb{P}_\kappa$  be the free filters on  $\omega$  generated, modulo the cofinite filter, by fewer than  $\kappa$  sets. Its Boolean completion  $\mathbb{C}_\kappa$  is the regular-open algebra of the topology on  $\omega^*$  generated by the ultrafilter-extension sets of these filters. The original  $\mathcal{P}(\omega)/\text{Fin}$  remains order dense exactly for  $\kappa \leq \mathfrak{p}$ . Above this threshold there is no complete Boolean homomorphism from the original completion to  $\mathbb{C}_\kappa$  fixing all original events, and none from  $\mathbb{C}_\kappa$  back to the original completion either. The distributivity number  $\mathfrak{h}$  is the least size of a family of completed binary events of the original completion that cannot be decided jointly on a dense set of regions, and at the successor stages above  $\mathfrak{h}$  the completion changes even abstractly. Atoms are exactly ultrafilters of character below  $\kappa$ , so they first occur at  $\mathfrak{u}^+$ . We characterize full atomicity, describe the residual part as the region where forcing adds a new ultrafilter, and derive a three-stage example from Millán's published model. The reduced-power sheafification is constructed by Boolean mixing. Finitary first-order truth is preserved and existential values are attained, and at the successor stages every ultrafilter specialization is countably saturated; a  $\mathfrak{p}$ -fold conjunction of fixed old predicates separates the completions sharply. Global rings over fields are von Neumann regular, with the completion as their idempotent algebra. For  $2 \leq |M| \leq \mathfrak{c}$  the raw global reduced power, the countably generated sheafification and the full-site sheafification have  $\mathfrak{c}$ ,  $2^{\mathfrak{c}}$  and  $2^{2^{\mathfrak{c}}}$  elements. A final program collects the questions of both manuscripts, re-scoped where one answers the other.

## Contents

|          |                                                                          |           |
|----------|--------------------------------------------------------------------------|-----------|
| <b>1</b> | <b>The repository question and the contribution</b>                      | <b>4</b>  |
| 1.1      | A gap left by the existing gluing analysis . . . . .                     | 4         |
| 1.2      | The non-claim of the earlier report that this report continues . . . . . | 4         |
| 1.3      | Questions answered and what is classical . . . . .                       | 5         |
| 1.4      | A map of the principal conclusions . . . . .                             | 6         |
| 1.5      | The two sources and how they are merged . . . . .                        | 6         |
| 1.6      | Conventions and the notation dictionary . . . . .                        | 6         |
| <b>2</b> | <b>Filters, generating character, and dense covers</b>                   | <b>8</b>  |
| <b>3</b> | <b>A concrete representation of the Booleanization</b>                   | <b>9</b>  |
| 3.1      | A topology on the set of free ultrafilters . . . . .                     | 9         |
| 3.2      | Regular opens and the dense topology . . . . .                           | 10        |
| 3.3      | The poset-side description . . . . .                                     | 11        |
| <b>4</b> | <b>The exact pseudointersection threshold</b>                            | <b>12</b> |
| 4.1      | The two cardinal invariants . . . . .                                    | 12        |
| 4.2      | A general completion criterion . . . . .                                 | 13        |
| 4.3      | Preservation and its sharp failure . . . . .                             | 14        |
| 4.4      | An exact arity invariant and an explicit inverse map . . . . .           | 16        |

|           |                                                                                 |           |
|-----------|---------------------------------------------------------------------------------|-----------|
| <b>5</b>  | <b>Closure and simultaneous Boolean refinement</b>                              | <b>17</b> |
| 5.1       | At which stages the closure consequences are proved . . . . .                   | 18        |
| <b>6</b>  | <b>The distributivity number as a local-reading boundary</b>                    | <b>18</b> |
| <b>7</b>  | <b>The ultrafilter threshold: atoms, points, and atomicity</b>                  | <b>19</b> |
| 7.1       | The exact atom spectrum . . . . .                                               | 19        |
| 7.2       | Full atomicity is an extension property . . . . .                               | 21        |
| <b>8</b>  | <b>A forcing interpretation of the residual part</b>                            | <b>22</b> |
| <b>9</b>  | <b>Early atomicity and genuinely different atomic stages</b>                    | <b>23</b> |
| <b>10</b> | <b>A countable toy model and the marking issue</b>                              | <b>24</b> |
| <b>11</b> | <b>Reduced powers and their explicit mixing sheaves</b>                         | <b>25</b> |
| 11.1      | The raw presheaf is separated . . . . .                                         | 25        |
| 11.2      | Mixtures over a Boolean element . . . . .                                       | 25        |
| 11.3      | The associated-sheaf theorem . . . . .                                          | 27        |
| <b>12</b> | <b>First-order transfer, specializations, and a sharp infinitary separation</b> | <b>29</b> |
| 12.1      | Boolean truth on mixtures . . . . .                                             | 29        |
| 12.2      | Ordinary specializations . . . . .                                              | 30        |
| 12.3      | An explicit infinitary formula detecting the threshold . . . . .                | 31        |
| 12.4      | The stable range includes the whole sheaf model . . . . .                       | 31        |
| <b>13</b> | <b>Localized realization and saturated specializations</b>                      | <b>31</b> |
| 13.1      | Scope of the model-theoretic conclusion . . . . .                               | 33        |
| <b>14</b> | <b>Global rings: regularity and the idempotent spectrum</b>                     | <b>33</b> |
| <b>15</b> | <b>The two endpoints and their exact cardinalities</b>                          | <b>35</b> |
| <b>16</b> | <b>General Boolean algebras and intermediate comparisons</b>                    | <b>36</b> |
| <b>17</b> | <b>A consolidated picture and worked regimes</b>                                | <b>37</b> |
| 17.1      | Under the continuum hypothesis . . . . .                                        | 37        |
| 17.2      | When the characteristics separate . . . . .                                     | 37        |
| 17.3      | Why ordinary model theory does not see everything . . . . .                     | 37        |
| <b>18</b> | <b>Proof audit and reproducible finite checks</b>                               | <b>38</b> |
| 18.1      | Foundational and logical boundaries . . . . .                                   | 38        |
| 18.2      | Dependencies of the principal results . . . . .                                 | 39        |
| 18.3      | What the executable supplement checks . . . . .                                 | 39        |

|                                                            |           |
|------------------------------------------------------------|-----------|
| 18.4 A formalization path . . . . .                        | 40        |
| <b>19 Further research questions and proposed projects</b> | <b>40</b> |
| 19.1 Source 08's questions . . . . .                       | 41        |
| 19.2 Source 10's questions . . . . .                       | 42        |
| <b>20 Conclusion</b>                                       | <b>44</b> |
| <b>A Provenance and the merge record</b>                   | <b>45</b> |
| A.1 The two manuscripts . . . . .                          | 45        |
| A.2 Choices made by the merge . . . . .                    | 46        |
| A.3 Checks made at intake and in the write . . . . .       | 46        |
| A.4 Concordance of numbers . . . . .                       | 47        |

# 1 The repository question and the contribution

## 1.1 A gap left by the existing gluing analysis

Massas’s possibility-semantic construction of the hyperreal line uses reduced powers indexed by proper filters extending the cofinite filter. His published question after Fact 5.10 asks whether the uncountability assumption in his failure-of-sheafness argument is necessary [2, p. 517]. The ProveIt manuscript *A two-valued obstruction to gluing Fréchet powers* gives a negative answer using two constants, and also proves failure on the countably generated subsite [1]. We do not claim that resolution again. In Massas’s setting the full filter poset and its regular-open algebra are already part of the literature [2, Sections 5.2 and 6.1]. The repository report is unrefereed and is not a Lean theorem, and no result below is justified merely by its presence in that report.

There is, however, an important distinction between the two associated sheaves. On the full filter site, maximal filters form a dense family; in ZFC the associated sheaf has values

$$F \mapsto \prod_{U \supseteq F} M^\omega / U.$$

On the countably generated free-filter site there are no ultrafilter objects. Reusing the full-site formula on that subsite would give the wrong answer. Our starting question is thus precise:

What is the Booleanization of the smaller filter site, and exactly how much filter complexity must be admitted before its truth values and its points change?

We answer this question and compute the corresponding reduced-power sheafification. The two-valued obstruction is background motivation, not an unproved premise of our results.

## 1.2 The non-claim of the earlier report that this report continues

The earlier report ends its hyperreal discussion with a limitation of scope [1, Section 10]:

“The result also depends on the category and its topology. It is not an assertion that every sheaf-based construction of nonstandard analysis fails, or that every reduced-power presheaf on a different filter site has the same behavior. The relative countably generated site considered in Section 7 was analyzed separately; no unexamined change of topology is implicit in the main theorem.”

This report continues exactly that non-claim: it examines the reduced-power presheaf on a whole cardinal-indexed family of other filter sites, the bounded-character sites  $\mathbb{P}_\kappa$  of Section 2, which interpolate between the countably generated site of that report’s Section 7 ( $\mathbb{P}_{\aleph_1}$ ) and its full site ( $\mathbb{P}_{c^+}$ ).

It does *not* re-claim that report’s results. Its two-valued nongluing theorem (its Theorem 1.1), the failure on the countably generated site (its Section 7.1), separatedness (its Theorem 5.1), finite pasting (its Theorem 5.2), the local finite-quotient classification (its Theorem 8.5) and the full-site product sheafification (its Theorem 9.2) are that report’s. Where the present text reproves one of them (separatedness in Proposition 11.1, the full-site product in Corollary 11.10), it does so only to connect it to the uniform construction here. Neither source manuscript answers the one question that report names, whether the infinite-quotient direction of its Theorem 8.5 needs dependent choice; see question R8 in Section 19.

### 1.3 Questions answered and what is classical

There are several genuinely different meanings of “complete the Fréchet power.” The raw binary reduced power is  $\mathcal{P}(\omega)/\text{Fin}$ . Its order-dense Boolean completion is atomless. The full filter-site completion is instead a power-set algebra, hence atomic. This raises an interpolation problem: how much filter complexity is needed to pass from one behavior to the other? The report answers three questions about it. First, exactly when does the original modulo-finite algebra remain order dense in the completed truth-value algebra, and when does every complete comparison fixing the original events fail? Second, when do complete Boolean points appear, and what part of the completion do they see? Third, how large a family of binary sections can be decided simultaneously on a dense collection of local regions? The answers involve  $\mathfrak{p}$ ,  $\mathfrak{u}$  and  $\mathfrak{h}$ , respectively. A change in the complete algebra of truth values need not destroy transfer or saturation, and the model-theoretic sections show that at the successor stages it does not: the mixed Fréchet models keep finitary transfer, attained existential values and a localized countable realization property.

The pseudointersection number  $\mathfrak{p}$ , the ultrafilter number  $\mathfrak{u}$ , the distributivity number  $\mathfrak{h}$ , and their basic filter interpretations are classical [3]. Likewise, Stone duality, Boolean completion, closure-to-distributivity arguments, the countable diagonal, and the passage from Boolean-valued models to sheaves by mixing are established tools. The recent treatment of Pierobon and Viale explicitly develops the relationship between mixing and dense-topology sheaves [4], and Ulrich’s treatment of full Boolean-valued models covers their ordinary ultrafilter specializations [6, Sections 2–3]. We give the specialized proofs needed here so that no semantic convention is left implicit.

The contribution is a single, explicitly marked hierarchy connecting these tools to the repository’s filter site: an exact obstruction to complete event-preserving comparison, in both directions; the complete atom and point spectrum; an early-atomicity application; a local-reading interpretation of  $\mathfrak{h}$ ; a concrete sheafification; a sharp infinitary truth-value gap; countable saturation of specializations; and exact cardinalities of the endpoints. The cardinal thresholds are translations of classical combinatorics, not newly discovered cardinal invariants. Recovering  $\mathfrak{p} \leq \mathfrak{h}$  (Corollary 6.3) is a consistency check, not a new inequality. The full-site power-set identification is already discussed by Massas, and the full-site product sheaf is already in the repository report. The resulting statements are research extensions of the inspected report; targeted literature searches by both source manuscripts did not locate this exact bounded-character package, but that is not an exhaustive novelty or priority determination.

**Main distinction.** The same finite Boolean algebraic events can sit inside different complete algebras. Preserving the old events and preserving arbitrary joins are separate requirements. The first does not force the second, and above  $\mathfrak{p}$  the two requirements become incompatible with either canonical direction of comparison.

The repository snapshot of both manuscripts is commit `e21766d04c2b8a9b2cdba0cd43e563024b3bd1b9`. The relevant report is in `SetTheory/Cardinals/docs/reports/ordinals-and-order-types/frechet-two-valued-obstruction/`. Its text was unchanged from that snapshot to the placement of this report; the only later change is a remark pointing here, added when this report was written. Its own documentation labels it unrefereed and not formally verified. We therefore give independent proofs of the repository-specific structural facts used below instead of inheriting a claim of verification.

## 1.4 A map of the principal conclusions

For an infinite cardinal  $\kappa$ , let  $\mathbb{C}_\kappa$  denote the completion defined in Section 3. The following table records the results, with proofs later in the report.

| Question                                                                                                                                                      | Exact answer                                                                                             |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| When does $\mathbb{C}_\kappa$ retain the original completion over $\mathcal{P}(\omega)/\text{Fin}$ ?                                                          | Exactly when $\kappa \leq \mathfrak{p}$ .                                                                |
| When is there a complete map fixing the original events, from $\mathbb{C}_\omega$ to $\mathbb{C}_\kappa$ or from $\mathbb{C}_\kappa$ to $\mathbb{C}_\omega$ ? | In either direction, exactly when $\kappa \leq \mathfrak{p}$ .                                           |
| How many completed binary events of $\mathbb{C}_\omega$ can be decided jointly on a dense set?                                                                | Fewer than $\mathfrak{h}$ , and not always $\mathfrak{h}$ .                                              |
| When does $\mathbb{C}_\kappa$ have no atoms or geometric points?                                                                                              | Exactly when $\kappa \leq \mathfrak{u}$ .                                                                |
| When is it completely atomic?                                                                                                                                 | Every allowed filter extends an allowed ultrafilter.                                                     |
| What sheaf replaces $M^\omega/F$ ?                                                                                                                            | Arbitrary Boolean mixtures of old sequences.                                                             |
| When can a zero meet of old events become positive?                                                                                                           | Above $\mathfrak{p}$ , first at arity $\mathfrak{p}$ .                                                   |
| Are ultrafilter specializations saturated?                                                                                                                    | Countably, at every successor stage.                                                                     |
| What happens for fields?                                                                                                                                      | A regular global ring with idempotent algebra $\mathbb{C}_\kappa$ .                                      |
| How large are the global sections for $2 \leq  M  \leq \mathfrak{c}$ ?                                                                                        | $\mathfrak{c}$ raw, $2^\mathfrak{c}$ at $\mathbb{P}_{\aleph_1}$ , $2^{2^\mathfrak{c}}$ at the full site. |

Here and throughout, “over the original Boolean algebra” means *fixing its specified embedding*. Abstract isomorphism is a different question, and Section 10 gives an explicit reason not to confuse them.

## 1.5 The two sources and how they are merged

This report is built from two manuscripts of batch 36 of ProveIt’s incoming-reports intake, both dated 24 September 2026, both pinned to the snapshot above, and written without knowledge of each other:

- **Source 08**, *Filter Complexity and Booleanization: Pseudointersection thresholds, ultrafilter atoms, and Fréchet-power sheaves* (24 pages). It is the *base*: its site index  $\mathbb{P}_\kappa$  is strictly more general than source 10’s, and its development is kept in full.
- **Source 10**, *Three Cardinal Boundaries for Fréchet-Power Completions: Pseudointersections, local decisions, and ultrafilter points* (26 pages). Its unique results are integrated: the reverse comparison barrier, the  $\mathfrak{h}$  boundary, the closure and partition-refinement consequences, the forcing reading, the specialization and saturation theorems, and the exact sizes.

Every result, proof, example, remark, limitation and question of both sources is printed. A result proved by both is printed once, and its heading names both sources; a genuinely different second proof is kept as a marked second route. A heading such as [08 Thm 4.2; 10 Thm 4.3(a)(b)] names the numbers of the delivered PDFs; Appendix A.4 maps them to the numbers here. Paragraphs written for the merge, not taken from either source, are set as *Merge notes*. Appendix A records the provenance and every choice the merge had to make.

## 1.6 Conventions and the notation dictionary

The two sources index their stages differently, and this is the one point where a careless reading gives wrong thresholds. We fix *source 08*’s convention throughout.

- Source 08 bounds the number of generators modulo the cofinite filter *strictly*:  $\mathbb{P}_\kappa = \{F : \chi_*(F) < \kappa\}$ .

- Source 10 bounds the ordinary base character *weakly*: its site is  $\mathcal{F}_\lambda = \{F : \chi(F) \leq \lambda\}$  for infinite  $\lambda$ .
- Lemma 2.2 proves  $\mathcal{F}_\lambda = \mathbb{P}_{\lambda^+}$  for every infinite  $\lambda$ . So source 10's stages are exactly the *successor* stages here. Source 08 has further stages: the finite-generation stage  $\mathbb{P}_\omega$  and the limit stages  $\mathbb{P}_\kappa$  ( $\kappa$  a limit cardinal), for example  $\mathbb{P}_{\aleph_\omega}$ , which is no  $\mathcal{F}_\lambda$ .

Every statement imported from source 10 is translated. The table lists the translations and the renamed symbols.

| Source 10                                                                                      | This report (= source 08)                                                                      | Note                                                                              |
|------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| $\chi(F)$ , base character                                                                     | $\chi(F)$ ; source 08's $\chi_*(F)$ counts generators mod Fr                                   | $\chi_* \leq \chi \leq \max(\chi_*, \aleph_0)$                                    |
| $\mathcal{F}_\lambda$ ( $\chi \leq \lambda$ )                                                  | $\mathbb{P}_{\lambda^+}$ ( $\chi_* < \lambda^+$ )                                              | $\mathcal{F}_{\aleph_0} = \mathbb{P}_{\aleph_1}$ , $\mathcal{F}_c = \mathbb{P}_c$ |
| $\mathbb{D}_\lambda = \text{RO}(\mathcal{F}_\lambda)$                                          | $\mathbb{C}_{\lambda^+}$                                                                       | Lemma 3.6                                                                         |
| $\mathbb{C}$ , the one completion of $\mathbb{B}$                                              | $\mathbb{C}_\omega$ ( $= \text{RO}(\omega^*)$ , marked-isomorphic to $\mathbb{C}_{\aleph_1}$ ) | a bare $\mathbb{C}$ is never used here                                            |
| $j_\lambda : \mathbb{B} \rightarrow \mathbb{D}_\lambda; i : \mathbb{B} \rightarrow \mathbb{C}$ | $e_{\lambda^+}; e_\omega$                                                                      |                                                                                   |
| cone $\langle F \rangle$                                                                       | $\uparrow F$ in the poset, $K_F$ in $\omega^*$                                                 | $\langle \cdot \rangle$ is the <i>generated filter</i> here                       |
| $\text{fil}(\cdot)$ , generated filter                                                         | $\langle \cdot \rangle$                                                                        | as in the earlier report                                                          |
| $\lambda < \mathfrak{p}$                                                                       | $\kappa = \lambda^+ \leq \mathfrak{p}$                                                         | marked completion survives                                                        |
| $\lambda \geq \mathfrak{p}$                                                                    | $\kappa = \lambda^+ > \mathfrak{p}$                                                            | barrier; <i>not</i> $\kappa \geq \mathfrak{p}$                                    |
| $\lambda < \mathfrak{u}$ ; atoms “at $\mathfrak{u}$ ”                                          | $\lambda^+ \leq \mathfrak{u}$ ; first atoms at $\mathbb{P}_{\mathfrak{u}^+}$                   |                                                                                   |
| $\lambda \geq \mathfrak{h}$                                                                    | $\kappa = \lambda^+ > \mathfrak{h}$                                                            | abstract non-isomorphism                                                          |
| $\mathcal{U}; \mathcal{U}_\lambda$ ( $\chi \leq \lambda$ )                                     | $\omega^*; S_{\lambda^+}$ ( $\chi_* < \lambda^+$ )                                             | $S_\kappa$ is a set of ultrafilters                                               |
| $a_\lambda, r_\lambda = \neg a_\lambda$                                                        | $a_{\lambda^+}, r_{\lambda^+} = \neg a_{\lambda^+}$                                            | same meaning                                                                      |
| admissible site $P, \mathbb{D}_P, j_P$                                                         | $Q, \text{RO}(Q), e_Q$                                                                         | $\mathbb{P}$ is reserved for the stages                                           |
| $J, H$ (comparison maps)                                                                       | $J, H$ ; source 08's $h$ is renamed $J$                                                        | $\mathfrak{h}$ is the distributivity number                                       |
| $E_F$ (dense incompatible family)                                                              | $\mathcal{E}_F$                                                                                | $E(f, g)$ is an equality set                                                      |
| $D_b, D_\xi$ (dense sets in $\mathbb{B}^+$ )                                                   | $\mathcal{D}_b, \mathcal{D}_\xi$                                                               | $D$ is a sieve or a subset of $\omega$                                            |
| $e_\xi$ (real codes)                                                                           | $\gamma_\xi$                                                                                   | $e_\kappa$ is the embedding                                                       |
| value set $A$ , structure $M$                                                                  | $M$ for both                                                                                   | $A$ is a subset of $\omega$                                                       |
| $\text{Mix}_i(b_i, f_i)$                                                                       | $\sum_i b_i \cdot f_i$                                                                         | a presentation                                                                    |
| $S_\lambda(A); S_\lambda(A; d)$                                                                | $\text{Mix}_M(1); \text{Mix}_M(w)$ at stage $\lambda^+$                                        | $S_\kappa$ is a set of ultrafilters here                                          |
| $\mathcal{S}_{\lambda, A}$                                                                     | $\mathcal{S}_{M, \lambda^+}$                                                                   | $\mathcal{S}_{M, c^+}$ is the earlier report's $\mathcal{S}_A$                    |
| $\ \cdot\ ; T_\varphi$                                                                         | $\ \cdot\ ; V_\varphi$                                                                         |                                                                                   |
| region $d; d_m, d_\infty; e_m$                                                                 | $w; \pi_m, \pi_\infty; \theta_m$                                                               | $d_\kappa$ is the arity invariant                                                 |
| $E_m$ (satisfiability sets)                                                                    | $\Sigma_m$                                                                                     |                                                                                   |
| ultrafilter $V$ on $\mathbb{D}_\lambda$ ; ground model $V$                                     | $\mathcal{V}$ on $\mathbb{C}_{\lambda^+}$ ; ground model $\mathcal{V}$                         | $V$ is a regular open, $V_\varphi$ a truth set                                    |
| $\Omega$ (independent-family index)                                                            | $I$                                                                                            | same construction                                                                 |
| field (source 08's $K$ )                                                                       | $\mathbb{F}$                                                                                   | $K_F$ is an ultrafilter-extension set                                             |

Two tempting false readings deserve a separate warning.

- “*The barrier starts at  $\kappa = \mathfrak{p}$ .*” False here:  $\mathbb{P}_\mathfrak{p}$  admits only filters with fewer than  $\mathfrak{p}$  generators, and the marked completion is still the original one at  $\kappa = \mathfrak{p}$ . Source 10's “ $\lambda \geq \mathfrak{p}$ ” means  $\kappa = \lambda^+ > \mathfrak{p}$ . Likewise source 10's “atoms first appear at  $\mathfrak{u}$ ” and source 08's “the first stage with atoms is  $\mathfrak{u}^+$ ” say the same thing in the two indexings.
- “*Source 10's closure consequences hold at every stage.*” They are proved at source 10's stages, the successor stages  $\mathbb{P}_{\lambda^+}$ . At a singular stage of countable cofinality such as  $\mathbb{P}_{\aleph_\omega}$  the union of a countable chain can need  $\aleph_\omega$  generators and lie outside the site; those results are *not* proved there (Section 5.1).

Theorem-environment numbers in the delivered PDFs coincide across the sources but denote different results (“Theorem 4.3” is source 10’s barrier and “Corollary 4.3” source 08’s countable site), so a source number is always prefixed by its source.

## 2 Filters, generating character, and dense covers

All arguments in this report take place in ZFC; all sets and all cardinalities are external. Put  $\mathfrak{c} = 2^{\aleph_0}$  and

$$\text{Fr} = \{A \subseteq \omega : \omega \setminus A \text{ is finite}\}, \quad \mathbb{B} = \mathcal{P}(\omega)/\text{Fin}.$$

A free filter means a proper filter containing Fr; in particular it has no finite member. The notation  $A \subseteq^* D$  means that  $A \setminus D$  is finite, and  $[A]$  is the class of  $A$  in  $\mathbb{B}$ . Boolean order on  $\mathbb{B}$  is almost inclusion. A complete Boolean homomorphism is always unital and preserves *all* joins; it therefore preserves complements and all meets.

**Definition 2.1** (Two characters [08 Def. 2.1; 10 §2.1]). For a free filter  $F$ , set

$$\chi_*(F) = \min\{|\mathcal{A}| : F = \langle \text{Fr} \cup \mathcal{A} \rangle\},$$

the generating character modulo finite sets (source 08), and let  $\chi(F)$  be the least cardinality of a base  $\mathcal{A} \subseteq F$ , meaning that every member of  $F$  contains a member of  $\mathcal{A}$  (source 10’s ordinary base character, not a  $\pi$ -character). For an infinite cardinal  $\kappa$  define

$$\mathbb{P}_\kappa = \{F : F \text{ is a free filter and } \chi_*(F) < \kappa\}.$$

The order is the forcing order:  $G \leq F$  means  $G \supseteq F$ , and a *stronger* condition is a larger filter. Equivalently, the category has an arrow  $G \rightarrow F$  for every such inclusion.

The use of  $\chi_*$  is deliberate. The cofinite filter has  $\chi_*(\text{Fr}) = 0$ , although its usual inclusion-base character is countably infinite. For a nonprincipal ultrafilter the two characters coincide: closing a family under finite intersections and deleting finite sets multiplies an infinite size only by  $\aleph_0$ , while no nonprincipal ultrafilter has a countable base. The latter fact also follows from Lemma 4.1 below. The next lemma makes the comparison exact; it is the dictionary between the two sources’ indexings.

**Lemma 2.2** (The index shift [merge]). *For every free filter  $F$ ,*

$$\chi_*(F) \leq \chi(F) \leq \max(\chi_*(F), \aleph_0), \quad \chi(F) \geq \aleph_0.$$

*Consequently, for every infinite cardinal  $\lambda$ , source 10’s site  $\mathcal{F}_\lambda = \{F : \chi(F) \leq \lambda\}$  equals  $\mathbb{P}_{\lambda^+}$ . For a free ultrafilter  $U$ ,  $\chi(U) = \chi_*(U)$ , so the ultrafilter number  $\mathfrak{u}$  is the same whether defined with  $\chi$  (source 10) or with  $\chi_*$  (source 08).*

*Proof.* A base of  $F$  generates  $F$ , so  $\chi_*(F) \leq \chi(F)$ . Conversely, if  $\mathcal{A}$  generates  $F$  modulo Fr, every member of  $F$  contains a set  $A_1 \cap \dots \cap A_m \cap C$  with  $A_i \in \mathcal{A}$  and  $C$  cofinite, hence contains  $(A_1 \cap \dots \cap A_m) \setminus n$  for some  $n < \omega$ . These sets lie in  $F$  and form a base of size at most  $\max(|\mathcal{A}|, \aleph_0)$ . If  $F$  had a finite base, the intersection  $B$  of its members would lie in  $F$  and be infinite; then  $B \setminus \{\min B\} \in F$  would contain no member of the base, since each contains  $B$ . Hence  $\chi(F) \geq \aleph_0$ . For infinite  $\lambda$  the three inequalities give  $\chi(F) \leq \lambda \iff \chi_*(F) \leq \lambda \iff \chi_*(F) < \lambda^+$ . For a free ultrafilter,  $\chi(U)$  is uncountable (Lemma 4.1), so  $\chi_*(U)$  is uncountable too and the bounds coincide.  $\square$

**Lemma 2.3** (The finite-generation endpoint [08 Lemma 2.2; 10 eq. (1)]). *For every infinite  $A \subseteq \omega$ , let*

$$F_A = \{D \subseteq \omega : A \subseteq^* D\}.$$

*Then  $\mathbb{P}_\omega$  consists exactly of the filters  $F_A$ , and  $[A] \mapsto F_A$  is an order isomorphism from  $\mathbb{B}^+$  onto  $\mathbb{P}_\omega$ . For any free filter  $F$ ,*

$$F_A \supseteq F \iff A \subseteq^* D \text{ for every } D \in F.$$

*Each  $F_A$  has the countable base  $A \setminus n$  ( $n < \omega$ ), and  $F_{A'} \supseteq F_A$  exactly when  $A' \subseteq^* A$ .*

*Proof.* A finite generating family can be replaced by its intersection  $A$ . Properness implies that  $A$  is infinite, and adjoining the cofinite sets gives exactly  $F_A$ . Moreover,  $F_A \supseteq F_D$  precisely when  $A \subseteq^* D$ . The equivalence is just the definition of  $F_A$ , and the base is displayed by the definition.  $\square$

Source 10 calls the filters  $F_A$  the principal-modulo-finite subsite. “Principal” is meant modulo finite sets:  $F_A$  is a free filter, not an ordinary principal ultrafilter. For  $\kappa = \omega_1$ , the objects of  $\mathbb{P}_\kappa$  are precisely the countably generated free filters. Every free filter has at most  $\mathfrak{c} = 2^{\aleph_0}$  generators, so  $\mathbb{P}_{\mathfrak{c}^+}$  is the full free-filter site, and  $\mathbb{P}_\kappa = \mathbb{P}_{\mathfrak{c}^+}$  for every  $\kappa > \mathfrak{c}$  (source 10:  $\mathcal{F}_\lambda = \mathcal{F}_\mathfrak{c}$  for  $\lambda \geq \mathfrak{c}$ ).

A sieve  $D$  at  $F$  is a family of extensions of  $F$  closed under further extension. It is *dense covering* if

$$\forall G \supseteq F (G \in \mathbb{P}_\kappa \Rightarrow \exists H \in D (H \supseteq G)). \quad (1)$$

The condition is relative to the allowed filters: the same family need not have the same covering behavior after the site is enlarged.

**Lemma 2.4** (Small extensions [08 Lemma 2.3; 10 Lemma 2.1]). *Let  $F$  be a free filter and  $A \subseteq \omega$ . The filter generated by  $F \cup \{A\}$  is proper exactly when  $A$  is  $F$ -positive, meaning  $\omega \setminus A \notin F$ . If  $F \in \mathbb{P}_\kappa$  and  $A$  is  $F$ -positive, then  $\langle F \cup \{A\} \rangle$  is a member of  $\mathbb{P}_\kappa$ . The filter generated by two compatible members of  $\mathbb{P}_\kappa$  is again a member. Thus compatibility can be checked by ordinary filter compatibility without leaving the site, and it does not depend on the bound once the two filters are admitted.*

*Proof.* An improper generated filter would contain an empty intersection  $B \cap A$  with  $B \in F$ , which is equivalent to  $\omega \setminus A \in F$ . Otherwise every member of  $F$  meets  $A$ ; since  $F$  contains all cofinite sets, every such intersection is in fact infinite. The number of generators increases by at most one. Likewise, a union of two generating families has size below  $\kappa$ , and its filter is proper exactly when the original filters are compatible. Only a finite union of small families is involved, so regularity of  $\kappa$  is not needed.  $\square$

### 3 A concrete representation of the Booleanization

#### 3.1 A topology on the set of free ultrafilters

Let  $X = \text{Ult}(\mathbb{B}) = \omega^*$ , with its Stone topology  $\tau$ . If  $b = [A] \in \mathbb{B}$ , write

$$\widehat{b} = \{U \in X : A \in U\}.$$

These sets are clopen, and they form a base for  $\tau$ . For a free filter  $F$  put

$$K_F = \{U \in X : F \subseteq U\}.$$

The ultrafilter lemma gives  $K_F \neq \emptyset$ .

**Definition 3.1** (The small-intersection topology [08 Def. 3.1]). Let  $\tau_\kappa$  be the topology on  $X$  generated by the sets  $K_F$ ,  $F \in \mathbb{P}_\kappa$ . Equivalently, it is generated by nonempty intersections of fewer than  $\kappa$  original Stone clopens. Put

$$\mathbb{C}_\kappa = \text{RO}(X, \tau_\kappa), \quad e_\kappa : \mathbb{B} \longrightarrow \mathbb{C}_\kappa, \quad e_\kappa(b) = \widehat{b}.$$

We keep this explicit convention instead of using an ambiguous  $G_\kappa$ -modification notation.

**Lemma 3.2** (Recovery and the clopen base [08 Lemma 3.2]). *For free filters  $F, G$  and  $A \subseteq \omega$ ,*

$$\begin{aligned} A \in F &\iff K_F \subseteq \widehat{[A]}, \\ G \supseteq F &\iff K_G \subseteq K_F. \end{aligned}$$

*The sets  $K_F$ ,  $F \in \mathbb{P}_\kappa$ , form a base of nonempty  $\tau_\kappa$ -clopens. Two such sets intersect exactly when their filters are compatible, and a nonempty intersection is the set corresponding to the generated joint filter.*

*Proof.* The forward implication in the first equivalence is immediate. If  $A \notin F$ , then  $F \cup \{\omega \setminus A\}$  generates a proper filter: otherwise some member of  $F$  would be contained in  $A$ . An ultrafilter extension of this filter belongs to  $K_F$  but not to  $\widehat{[A]}$ . This proves the equivalence and, by applying it to every member of  $F$ , proves the second one.

If  $\mathcal{A}$  generates  $F$  modulo Fr, then  $K_F = \bigcap_{A \in \mathcal{A}} \widehat{[A]}$ . It is closed in  $\tau$  and hence in the finer topology  $\tau_\kappa$ ; it is open by definition. Finite intersections remain of the same form, by Lemma 2.4. The compatibility assertion follows by extending the joint proper filter to an ultrafilter.  $\square$

## 3.2 Regular opens and the dense topology

For a topological space  $Y$ , an open set  $V$  is regular open when  $V = \text{int cl } V$ . Its regular-open algebra is a complete Boolean algebra, with operations

$$\bigvee_i V_i = \text{int cl } \bigcup_i V_i, \quad \bigwedge_i V_i = \text{int } \bigcap_i V_i, \quad \neg V = \text{int}(Y \setminus V). \quad (2)$$

For completeness, if  $W = \text{int } \bigcap_i V_i$  and each  $V_i$  is regular open, then  $\text{int cl } W \subseteq \text{int cl } V_i = V_i$  for every  $i$ . Being open,  $\text{int cl } W$  is therefore contained in  $W$ ; the reverse inclusion holds for every open  $W$ . This verifies the meet formula. The other formulas follow by the same regularization argument.

**Theorem 3.3** (Representation of the filter-site Booleanization [08 Thm 3.3]). *The map  $F \mapsto K_F$  identifies  $\mathbb{P}_\kappa$  with an order-dense subposet of  $\mathbb{C}_\kappa^+$ . The algebra  $\mathbb{C}_\kappa$  is its Boolean completion. A sieve  $D$  covers  $F$  exactly when*

$$\bigvee_{G \in D} K_G = K_F \quad \text{in } \mathbb{C}_\kappa. \quad (3)$$

*Consequently the dense-site sheaf category is*

$$\text{Sh}(\mathbb{P}_\kappa, J_{\text{dense}}) \simeq \text{Sh}(\mathbb{C}_\kappa),$$

*where the right side is the sheaf category of the complete Boolean algebra viewed as a locale, with covering families having join equal to the object covered.*

*Proof.* Lemma 3.2 gives the order embedding. Every nonempty regular open set contains a nonempty basic clopen  $K_F$ , proving order density. Separativity follows as well: if  $K_G \not\subseteq K_F$ , the nonempty open difference  $K_G \setminus K_F$  contains a condition incompatible with  $F$ .

Let  $V = \bigcup_{G \in D} K_G \subseteq K_F$ . If  $D$  is dense in the sense of (1), every nonempty basic open in  $K_F$  contains a member  $K_H$  with  $H \in D$ . Thus  $V$  is dense in  $K_F$ , which is clopen, and its regularization is  $K_F$ . Conversely, suppose  $V$  is dense in  $K_F$  and take  $H \supseteq F$ . Then  $K_H$  meets some  $K_G$  with  $G \in D$ . The joint filter is allowed and belongs to  $D$  because  $D$  is a sieve. It extends  $H$ , proving (1).

Finally, restriction from the Boolean algebra to its dense subposet preserves covers by (3). It also loses no sheaf data: every nonzero Boolean element is covered by the conditions below it, and a section over it is uniquely a matching family on that cover. This describes an inverse extension functor. The same matching-family construction extends morphisms, and restricting and extending in either order gives a natural isomorphism. This proves the claimed equivalence without assuming that the locale has points.  $\square$

In particular,  $\tau_\omega = \tau$  and  $\mathbb{C}_\omega = \text{RO}(\omega^*)$  is the ordinary order-dense completion of  $\mathbb{B}$ ; this is source 10's  $\mathbb{C}$ , which it also constructs as  $\text{RO}(\mathbb{B}^+)$ . At the opposite endpoint, every ultrafilter is allowed in  $\mathbb{P}_{c^+}$ , so every singleton  $K_U = \{U\}$  is open. Hence

$$\mathbb{C}_{c^+} = \mathcal{P}(\omega^*). \quad (4)$$

This power-set algebra is generally *not* the order-dense completion of the original  $\mathbb{B}$ .

### 3.3 The poset-side description

Source 10 works with the regular opens of the filter poset itself. An open subset of  $\mathbb{P}_\kappa$  is a family closed under stronger extensions. For an open  $V$ , put

$$\neg V = \{F : \text{no } G \supseteq F \text{ belongs to } V\}, \quad (5)$$

$$\neg\neg V = \{F : \text{for every } G \supseteq F \text{ there is } H \supseteq G \text{ in } V\}, \quad (6)$$

where all quantified filters belong to  $\mathbb{P}_\kappa$ . A regular open is an open fixed by  $\neg\neg$ . The regular opens form a complete Boolean algebra  $\text{RO}(\mathbb{P}_\kappa)$ : its order is inclusion, its complement is (5), its finite meet is intersection, and its arbitrary join is  $\neg\neg(\bigcup_i V_i)$ . In the extension topology arbitrary intersections of opens are open; for regular opens they are also regular, so arbitrary meets are intersections. A sieve at  $F$  covers  $F$  exactly when the condition in (6) holds at  $F$ , which is (1). The principal cone of  $F$  is

$$\uparrow F = \{G \in \mathbb{P}_\kappa : G \supseteq F\}.$$

**Lemma 3.4** (Cones and old events [10 Lemma 2.2]). *Every cone  $\uparrow F$  is a nonzero regular open. The cones are order dense in  $\text{RO}(\mathbb{P}_\kappa)$ : every nonzero regular open contains one. The map*

$$j_\kappa : \mathbb{B} \longrightarrow \text{RO}(\mathbb{P}_\kappa), \quad j_\kappa([A]) = \{F : A \in F\}, \quad (7)$$

*is an injective Boolean homomorphism. For infinite  $A$  its value is  $\uparrow F_A$ , and for every  $F$ ,*

$$\uparrow F \subseteq j_\kappa([A]) \iff A \in F. \quad (8)$$

*Proof.* If  $H$  does not extend  $F$ , choose  $A \in F \setminus H$ . By Lemma 2.4,  $H$  has a proper extension containing  $\omega \setminus A$ ; no further extension can extend  $F$ . Thus  $H$  is not in the double negation of  $\uparrow F$ . This proves regularity; nonzeroness follows from  $F \in \uparrow F$ . A nonempty open containing  $F$  contains its cone, which proves order density.

Membership in a free filter is unchanged by a finite symmetric difference. Also  $A \in F$  if and only if  $F \supseteq F_A$  when  $A$  is infinite. A filter lies in  $\neg j_\kappa([A])$  exactly when it admits no proper extension containing  $A$ , equivalently when  $\omega \setminus A \in F$ . Therefore  $\neg j_\kappa([A]) = j_\kappa([\omega \setminus A])$ . Intersections are preserved by the filter axioms; finite joins then follow from complements and intersections. If  $A$  is infinite,  $F_A$  witnesses that  $j_\kappa([A]) \neq 0$ , so the Boolean homomorphism is injective. Finally, evaluate inclusion of cones at  $F$  itself to obtain (8).  $\square$

Source 10 states this lemma for its stages; the proof uses only Lemma 2.4 and holds verbatim at every stage.

*Remark 3.5* (Joins are not unions [10 Remark 2.3]). A filter can contain  $A \cup B$  while containing neither  $A$  nor  $B$ . Consequently  $j_\kappa([A \cup B])$  need not be the literal union of the two membership events. It is their join after regularization. Forgetting this distinction would invalidate later completion comparisons.

**Lemma 3.6** (The two descriptions agree [merge]). *The maps*

$$\text{RO}(\mathbb{P}_\kappa) \ni V \mapsto \bigvee_{F \in V} K_F \in \mathbb{C}_\kappa, \quad \mathbb{C}_\kappa \ni b \mapsto \{F \in \mathbb{P}_\kappa : K_F \subseteq b\},$$

*are mutually inverse complete Boolean isomorphisms. They send  $\uparrow F$  to  $K_F$  and  $j_\kappa([A])$  to  $e_\kappa([A])$ . In particular source 10's  $\mathbb{D}_\lambda = \text{RO}(\mathcal{F}_\lambda)$ , with its embedding  $j_\lambda$ , is  $\mathbb{C}_{\lambda^+}$  with  $e_{\lambda^+}$ .*

*Proof.* Let  $b \in \mathbb{C}_\kappa$  and  $V_b = \{F : K_F \subseteq b\}$ . It is upward closed by Lemma 3.2. If  $F \notin V_b$ , then  $K_F \not\subseteq b$ ; since  $K_F$  is open and  $b$  is regular open,  $K_F \cap \neg b \neq \emptyset$ , so some  $K_G \subseteq K_F \cap \neg b$  with  $G \supseteq F$ , and no extension of  $G$  lies in  $V_b$ . Hence  $V_b$  is regular. Both maps are monotone. By order density (Theorem 3.3),  $b = \bigvee \{K_F : K_F \subseteq b\}$ . Conversely, if  $V$  is regular and  $K_F \subseteq \bigvee_{G \in V} K_G$ , then every  $K_H \subseteq K_F$  meets some  $K_G$  with  $G \in V$ , and the joint filter extends  $H$  and lies in  $V$ ; so  $F \in \neg \neg V = V$ . The maps are therefore inverse order isomorphisms, hence complete Boolean isomorphisms. The cone  $\uparrow F$  goes to  $\bigvee_{G \supseteq F} K_G = K_F$ . For infinite  $A$ ,  $j_\kappa([A]) = \uparrow F_A$  goes to  $K_{F_A} = \widehat{[A]} = e_\kappa([A])$ , and for finite  $A$  both sides are 0. The last sentence is Lemma 2.2.  $\square$

## 4 The exact pseudointersection threshold

### 4.1 The two cardinal invariants

A family  $\mathcal{A} \subseteq [\omega]^\omega$  has the strong finite intersection property, or SFIP, if every finite intersection is infinite. An infinite  $A$  is a pseudointersection of  $\mathcal{A}$  if  $A \subseteq^* D$  for every  $D \in \mathcal{A}$ . For a filter  $F$ ,  $A$  is a pseudointersection of  $F$  exactly when  $F \subseteq F_A$ . Define

$$\begin{aligned} \mathfrak{p} &= \min\{|\mathcal{A}| : \mathcal{A} \text{ has SFIP and no infinite pseudointersection}\}, \\ \mathfrak{u} &= \min\{\chi_*(U) : U \text{ is a free ultrafilter}\}. \end{aligned}$$

These are the usual cardinal characteristics [3]. We use only these definitions, not any theorem identifying  $\mathfrak{p}$  with another cardinal characteristic; in particular, no result about the tower number is an input.

**Lemma 4.1** ([08 Lemma 4.1; 10 Lemmas 4.2 and 7.2]). *In ZFC,  $\aleph_1 \leq \mathfrak{p} \leq \mathfrak{u} \leq \mathfrak{c}$ . No free ultrafilter has an infinite pseudointersection. Every free filter generated modulo Fr by fewer than  $\mathfrak{p}$  sets has an infinite pseudointersection, and a family witnessing  $\mathfrak{p}$  generates a free filter  $F$  with  $\chi_*(F) \leq \mathfrak{p}$  and no infinite pseudointersection.*

*Proof.* For a countable SFIP family  $D_0, D_1, \dots$ , recursively choose a strictly increasing  $a_n \in \bigcap_{i \leq n} D_i$ . The set of the  $a_n$  is an infinite pseudointersection. Thus  $\mathfrak{p} > \aleph_0$ .

No free ultrafilter has an infinite pseudointersection. Indeed, if  $A$  were such a pseudointersection, split it into two infinite pieces  $A_0, A_1$ . If  $A_0 \in U$ , the inclusion  $A \subseteq^* A_0$  fails; if  $A_0 \notin U$ , then  $\omega \setminus A_0 \in U$ , and  $A \subseteq^* (\omega \setminus A_0)$  fails. A generating family for  $U$  therefore has no pseudointersection, proving  $\mathfrak{p} \leq \mathfrak{u}$ . Finally a free ultrafilter exists, and its size is at most  $\mathfrak{c}$ .

A generating family of a free filter has SFIP, and a pseudointersection of it is one of the generated filter, since  $F_A$  is a filter containing Fr. So fewer than  $\mathfrak{p}$  generators have a pseudointersection. Conversely, adjoining the cofinite sets to a witness for  $\mathfrak{p}$  preserves properness, and any pseudointersection of the resulting filter would be one of the witness family.  $\square$

## 4.2 A general completion criterion

The next theorem, from source 10, isolates the order-theoretic mechanism before any cardinal bound is imposed. An *admissible site*  $\mathcal{Q}$  is a set of free filters containing every  $F_A$  and closed under adjoining one set whenever the generated filter is proper. The constructions of Section 3.3 give  $\text{RO}(\mathcal{Q})$  and  $e_{\mathcal{Q}} : \mathbb{B} \rightarrow \text{RO}(\mathcal{Q})$ ,  $e_{\mathcal{Q}}([A]) = \{F \in \mathcal{Q} : A \in F\}$ , with the same proofs. Every  $\mathbb{P}_\kappa$  is admissible, by Lemmas 2.3 and 2.4.

**Theorem 4.2** (General completion criterion [10 Thm 3.2]). *For an admissible site  $\mathcal{Q}$ , the following are equivalent.*

- (i) *Every filter in  $\mathcal{Q}$  has an infinite pseudointersection.*
- (ii) *The filters  $F_A$  are dense in  $\mathcal{Q}$ .*
- (iii) *The subalgebra  $e_{\mathcal{Q}}(\mathbb{B})$  is order dense in  $\text{RO}(\mathcal{Q})$ .*
- (iv) *There is a complete Boolean homomorphism  $J : \mathbb{C}_\omega \rightarrow \text{RO}(\mathcal{Q})$  with  $J \circ e_\omega = e_{\mathcal{Q}}$ .*

*When these conditions hold, the homomorphism in (iv) is unique and is an isomorphism.*

*Proof.* The equivalence of (i) and (ii) is the identity  $F \subseteq F_A \iff A \subseteq^* D$  for every  $D \in F$ . If (ii) holds, every cone contains a cone  $\uparrow F_A = e_{\mathcal{Q}}([A])$ , proving (iii). Conversely, under (iii) choose  $0 < e_{\mathcal{Q}}([A]) \subseteq \uparrow F$ ; evaluating at  $F_A$  gives  $F \subseteq F_A$ , so (i) follows.

For (iii) $\Rightarrow$ (iv), here is the usual uniqueness-of-completion argument. For  $c \in \mathbb{C}_\omega$ , define

$$J(c) = \bigvee \{e_{\mathcal{Q}}(b) : b \in \mathbb{B}, e_\omega(b) \leq c\}. \quad (9)$$

The analogous construction in the opposite direction, using the order-dense copy of  $\mathbb{B}$  in  $\text{RO}(\mathcal{Q})$ , is inverse to  $J$ . To check this, it suffices to test the two compositions on the elements below each of the two dense copies. Equivalently, for  $b \in \mathbb{B}$ ,

$$e_\omega(b) \leq c \iff e_{\mathcal{Q}}(b) \leq J(c).$$

The forward implication is immediate. If the reverse failed, choose a nonzero  $a \in \mathbb{B}$  with  $e_\omega(a) \leq e_\omega(b) \wedge \neg c$  by density in  $\mathbb{C}_\omega$ . Then  $e_{\mathcal{Q}}(a)$  is nonzero and disjoint from every term of (9), contradicting  $e_{\mathcal{Q}}(b) \leq J(c)$ . The same argument on the other side proves that the two maps are inverse order isomorphisms. Order isomorphisms of complete Boolean algebras preserve all joins and complements. Uniqueness follows because  $c = \bigvee \{e_\omega(b) : e_\omega(b) \leq c\}$ .

It remains to prove (iv) $\Rightarrow$ (i); this direction contains the useful obstruction. Suppose  $F \in \mathcal{Q}$  has no pseudointersection, and put

$$\mathcal{E}_F = \{[A] \in \mathbb{B}^+ : \omega \setminus A \in F\}. \quad (10)$$

This set is dense in  $\mathbb{B}^+$ . Indeed, for any infinite  $Y$ , the failure of pseudointersection gives  $D \in F$  with  $Y \setminus D$  infinite. Set  $A = Y \setminus D$ . Then  $[A] \leq [Y]$  and  $\omega \setminus A \supseteq D$ , so  $[A] \in \mathcal{E}_F$ . Consequently  $\bigvee_{b \in \mathcal{E}_F} e_\omega(b) = 1$  in  $\mathbb{C}_\omega$ . But every  $e_{\mathcal{Q}}(b)$  with  $b \in \mathcal{E}_F$  is disjoint from the nonzero cone  $\uparrow F$ . A complete extension would therefore give

$$1 = J(1) = \bigvee_{b \in \mathcal{E}_F} e_{\mathcal{Q}}(b) \leq \neg \uparrow F < 1,$$

a contradiction. □

The obstruction is a failed preservation of a particular dense join. It does not merely say that the evident set-theoretic map is badly chosen: *no* complete extension of the prescribed old events can work.

### 4.3 Preservation and its sharp failure

**Theorem 4.3** (Exact marked-completion threshold [08 Thm 4.2; 10 Thm 4.3(a),(b)]). *For every infinite cardinal  $\kappa$ , the following are equivalent.*

- (i)  $\kappa \leq \mathfrak{p}$ .
- (ii) Every  $F \in \mathbb{P}_\kappa$  has an extension of the form  $F_A$ .
- (iii)  $e_\kappa[\mathbb{B}]$  is order dense in  $\mathbb{C}_\kappa$ .
- (iv) There is a complete Boolean isomorphism  $J : \mathbb{C}_\omega \rightarrow \mathbb{C}_\kappa$  satisfying  $Je_\omega = e_\kappa$ .
- (v) There is a complete Boolean homomorphism  $J : \mathbb{C}_\omega \rightarrow \mathbb{C}_\kappa$  satisfying  $Je_\omega = e_\kappa$ .

When the isomorphism in (iv) exists, it is unique.

*Proof.* Assume  $\kappa \leq \mathfrak{p}$ , and choose fewer than  $\kappa$  generators for  $F$ . They have SFIP and hence have an infinite pseudointersection  $A$ . Lemma 2.3 gives  $F_A \supseteq F$ , proving (ii). Conversely, such an extension is exactly a pseudointersection of the generators. If  $\kappa > \mathfrak{p}$ , a family witnessing  $\mathfrak{p}$  generates an allowed filter with no such extension. Thus (i) and (ii) are equivalent.

By Theorem 3.3, every positive Boolean element contains a  $K_F$ . Condition (ii) says that every such  $K_F$  contains an original nonempty clopen  $\widehat{[A]} = K_{F_A}$ , so (ii) implies (iii). Conversely, if  $\widehat{[A]} \subseteq K_F$ , Lemma 3.2 gives  $F_A \supseteq F$ . Hence (iii) implies (ii).

An order-dense Boolean embedding has a unique completion up to a complete isomorphism fixing the embedded algebra. One can see this directly here: each element of either completion is the join of the original Boolean elements below it; its set of such lower bounds is determined by which original Boolean elements are incompatible with it. The same incompatibility relation holds in both embeddings. Mapping these cuts to each other gives an order isomorphism and hence a complete Boolean isomorphism. This proves (iii) to (iv), including uniqueness. The implication from (iv) to (v) is immediate.

It remains to show that even (v) fails when  $\kappa > \mathfrak{p}$ . Let  $\mathcal{A}$  be an SFIP family of size  $\mathfrak{p}$  with no infinite pseudointersection, and let  $F = \langle \text{Fr} \cup \mathcal{A} \rangle$ . The set

$$K_F = \bigcap_{A \in \mathcal{A}} \widehat{[A]}$$

is nonempty and  $\tau_\kappa$ -clopen. Its original Stone interior is empty: a nonempty  $\widehat{[D]} \subseteq K_F$  would make  $D$  a pseudointersection. Formula (2) therefore gives

$$\bigwedge_{A \in \mathcal{A}} e_\omega([A]) = 0, \quad \bigwedge_{A \in \mathcal{A}} e_\kappa([A]) = K_F > 0. \quad (11)$$

A complete Boolean homomorphism fixing the original events would preserve these meets, which is impossible. □

*Second proof of the failure of (v)* [source 10]. For  $\kappa > \mathfrak{p}$ , Lemma 4.1 gives a filter in  $\mathbb{P}_\kappa$  without an infinite pseudointersection. By Lemma 3.6, a map  $J$  as in (v) is a map as in Theorem 4.2(iv) for  $Q = \mathbb{P}_\kappa$ , so it cannot

exist. The obstruction here is the dense *join* of the family  $\mathcal{E}_F$  of (10), not the  $\mathfrak{p}$ -fold *meet* of (11); the two are dual forms of one phenomenon, since  $\mathcal{E}_F$  consists of the old events disjoint from  $K_F$ .  $\square$

Source 08 proves only the direction from the old completion  $\mathbb{C}_\omega$  to  $\mathbb{C}_\kappa$ . Source 10 proves that the reverse direction fails as well.

**Theorem 4.4** (The reverse barrier [10 Thm 4.3(c)]). *If  $\kappa > \mathfrak{p}$ , there is no complete Boolean homomorphism  $H : \mathbb{C}_\kappa \rightarrow \mathbb{C}_\omega$  satisfying  $H \circ e_\kappa = e_\omega$ .*

*Proof.* Fix  $\kappa > \mathfrak{p}$  and let

$$N_\kappa = \{F \in \mathbb{P}_\kappa : F \text{ has no infinite pseudointersection}\}.$$

We claim that  $N_\kappa$  is dense. Given  $F$ , there is nothing to do when  $F \in N_\kappa$ . Otherwise choose an infinite pseudointersection  $A$ , so that  $F \subseteq F_A$ . Transport a fixed  $\mathfrak{p}$ -sized SFIP family without pseudointersection from  $\omega$  to  $A$  along a bijection. The filter generated by that family,  $A$ , and  $\text{Fr}$  extends  $F_A$ , needs at most  $\mathfrak{p} < \kappa$  generators modulo  $\text{Fr}$ , and has no pseudointersection. The last assertion follows because any such pseudointersection would be almost contained in  $A$  and would pull back to one of the original witness family. This proves the claim.

Now suppose  $H$  exists, and take  $F \in N_\kappa$ . For every  $A \in F$  we have  $K_F \leq e_\kappa([A])$  by Lemma 3.2, and hence

$$H(K_F) \leq \bigwedge_{A \in F} e_\omega([A]) = 0. \quad (12)$$

To justify the equality, a nonzero meet in  $\mathbb{C}_\omega$  would contain a nonzero  $e_\omega([X])$ , by order density. Then  $X \subseteq^* A$  for every  $A \in F$ , contrary to  $F \in N_\kappa$ . Since  $N_\kappa$  is dense,  $\bigvee_{F \in N_\kappa} K_F = 1$  in  $\mathbb{C}_\kappa$  (Theorem 3.3). Completeness of  $H$  and (12) give  $H(1) = 0$ , contradicting unitality.  $\square$

*Merge note.* Source 10 states Theorem 4.4 for its stages  $\mathcal{F}_\lambda$  with  $\lambda \geq \mathfrak{p}$ , that is, for successor  $\kappa = \lambda^+ > \mathfrak{p}$ . Its proof uses only the density of  $N_\kappa$ , and the extension built there needs at most  $\mathfrak{p}$  generators, which is allowed at every  $\kappa > \mathfrak{p}$ . No closure of the site is used. We therefore state it for every  $\kappa > \mathfrak{p}$ , limit stages included; this is the only place where a statement of source 10 is printed for stages it did not name, and the reason is this re-reading of the proof.

**Corollary 4.5** (The two-sided comparison boundary [10 Thm 4.3]). *For every infinite cardinal  $\kappa$ :*

- (a) *If  $\kappa \leq \mathfrak{p}$ , there is a unique complete Boolean isomorphism  $\mathbb{C}_\omega \cong \mathbb{C}_\kappa$  carrying  $e_\omega([A])$  to  $e_\kappa([A])$  for every  $A \subseteq \omega$ .*
- (b) *If  $\kappa > \mathfrak{p}$ , there is no complete Boolean homomorphism  $J : \mathbb{C}_\omega \rightarrow \mathbb{C}_\kappa$  satisfying  $J \circ e_\omega = e_\kappa$ .*
- (c) *If  $\kappa > \mathfrak{p}$ , there is also no complete Boolean homomorphism  $H : \mathbb{C}_\kappa \rightarrow \mathbb{C}_\omega$  satisfying  $H \circ e_\kappa = e_\omega$ .*

*Thus the existence of a complete comparison preserving the original modulo-finite events, in either direction, has the exact boundary  $\kappa \leq \mathfrak{p}$ . Parts (a) and (b) are in both sources; part (c), and hence the two-sided statement, is source 10's.*

*Proof.* Parts (a) and (b) are Theorem 4.3; part (c) is Theorem 4.4.  $\square$

**Corollary 4.6** (The countably generated site [08 Cor. 4.3; 10 eq. (9)]). *The dense Booleanization of the countably generated free-filter site is canonically  $\text{RO}(\omega^*)$  over  $\mathcal{P}(\omega)/\text{Fin}$ :*

$$\mathbb{C}_{\aleph_1} \cong \text{RO}((\mathcal{P}(\omega)/\text{Fin})^+) = \mathbb{C}_\omega. \quad (13)$$

In particular, it is not  $\mathcal{P}(\omega^*)$ : the former is atomless and the latter is atomic. More generally the marked completion is unchanged throughout  $\omega \leq \kappa \leq \mathfrak{p}$ ; its first possible change is at  $\mathfrak{p}^+$ .

*Proof.* Apply Theorem 4.3 and Lemma 4.1. The underlying reason is the elementary diagonal pseudointersection of a countable base. The algebra  $\mathbb{B}$  is atomless, because every infinite subset of  $\omega$  splits into two infinite pieces. Its order-dense completion is consequently atomless as well.  $\square$

The obstruction in (11) is stronger than failure of density. It rules out *any* complete event-preserving map in the displayed direction, not just an isomorphism, and Theorem 4.4 rules out the other direction too. Neither rules out an unmarked abstract isomorphism with a completely different action on the events  $[A]$ ; Corollary 6.3 gives an additional, unlabeled obstruction at the successor stages above  $\mathfrak{h}$ . The obstruction also does not assert that the identity on the underlying set of ultrafilters induces a morphism of Booleanized locales; it explains exactly why that assumption can fail.

Above  $\mathfrak{p}$ , the old algebra remains an injective Boolean subalgebra, but it stops being an order-dense subalgebra. In particular, “more filters” does not mean “a bigger complete algebra into which the old completion canonically embeds.” Nor does it produce a canonical complete quotient onto the old completion.

**Example 4.7** (A visible change in a dense cover [10 Example 4.4]). For a filter  $F \in \mathbb{P}_\kappa$  witnessing the failure above  $\mathfrak{p}$ , the family  $\mathcal{E}_F$  of (10) covers the weakest condition in the principal-modulo-finite site  $\mathbb{P}_\omega$ : its join in  $\mathbb{C}_\omega$  is 1. In the larger site it misses the entire nonzero region  $K_F$ . This one family of old binary events already witnesses failure of complete functoriality. The failure is therefore not confined to complicated model-theoretic formulas or infinite-valued sections.

#### 4.4 An exact arity invariant and an explicit inverse map

**Definition 4.8** ([08 Def. 4.4]). Let  $d_\kappa$  be the least size of a family  $\mathcal{A} \subseteq \mathbb{B}$  whose meet is zero in  $\mathbb{C}_\omega$  and positive in  $\mathbb{C}_\kappa$ , with  $d_\kappa = \infty$  if no such family exists. Both meets use the specified embeddings  $e_\omega, e_\kappa$ .

**Proposition 4.9** (The first zero-meet gap [08 Prop. 4.5]).

$$d_\kappa = \begin{cases} \infty, & \kappa \leq \mathfrak{p}, \\ \mathfrak{p}, & \kappa > \mathfrak{p}. \end{cases}$$

*Proof.* Below the threshold, the complete isomorphism preserves every meet. Above it, (11) gives  $d_\kappa \leq \mathfrak{p}$ . If a family of fewer than  $\mathfrak{p}$  old events has a positive meet in  $\mathbb{C}_\kappa$ , all of its finite meets are nonzero in  $\mathbb{B}$ . Choose subset representatives; they have SFIP. A pseudointersection gives a nonzero original Boolean lower bound, so the meet in  $\mathbb{C}_\omega$  cannot be zero. This gives the reverse inequality.  $\square$

**Proposition 4.10** (How a small filter is represented in the old completion [08 Prop. 4.6]). Assume  $\kappa \leq \mathfrak{p}$  and  $F \in \mathbb{P}_\kappa$ . Then

$$K_F = \text{cl}_\tau \text{int}_\tau K_F.$$

Under the inverse isomorphism  $\mathbb{C}_\kappa \rightarrow \mathbb{C}_\omega$  fixing  $\mathbb{B}$ , the element  $K_F$  is sent to  $\text{int}_\tau K_F$ .

*Proof.* Every original neighborhood of a point of  $K_F$  contains a clopen  $\widehat{[D]}$  meeting  $K_F$ . The filter generated by  $F$  and  $D$  has fewer than  $\mathfrak{p}$  generators, so it has a pseudointersection  $A$ . Then  $\widehat{[A]}$  lies in  $K_F \cap \widehat{[D]}$ . Thus the interior of  $K_F$  is dense in  $K_F$ , proving the first identity. The interior of a closed set is regular open. It is the join, in the old completion, of all original clopens contained in  $K_F$ , which is exactly the cut description of the inverse isomorphism.  $\square$

For the repository's row-tail filter, the countable intersection  $K_{F_\infty}$  (the earlier report's closed set  $K$ ) is closed but not originally clopen. It nevertheless has dense original interior, and that interior, not the raw subset  $K_{F_\infty}$ , is its representative in  $\text{RO}(\omega^*)$ . Thus the new calculation agrees with the old gluing obstruction without treating a closed set as if it were automatically a regular-open event.

## 5 Closure and simultaneous Boolean refinement

**Proposition 5.1** ([08 Prop. 5.5]). *If  $\kappa$  is regular and infinite, then  $\mathbb{P}_\kappa$  is  $< \kappa$ -closed: every descending forcing-order chain of length less than  $\kappa$  has a lower bound.*

*Proof.* In filter notation, such a chain is an increasing chain  $(F_\xi)_{\xi < \delta}$ , where  $\delta < \kappa$ . Its union is a proper filter: any finitely many of its members occur together at one stage, where their intersection is nonempty. Choose a small generating family at each stage. Regularity of  $\kappa$  ensures that the union of these  $\delta$  families has size below  $\kappa$ . It generates the union filter, which is the required lower bound. The empty chain is bounded by  $\text{Fr}$ .  $\square$

In particular, the countable-base presentation is countably closed. This assertion concerns a presentation, not a classification of all its Boolean-algebra invariants. For singular  $\kappa$ , the union of fewer than  $\kappa$  small bases can have size  $\kappa$ , so the proof does not apply. No singular closure conclusion is being smuggled into the threshold theorems, which did not need regularity.

Source 10 proves the same union argument at its stages and draws consequences from it. Every successor cardinal is regular, so at  $\kappa = \lambda^+$  Proposition 5.1 is exactly the first half of the next theorem; source 10 remarks that the argument needs no regularity of  $\lambda$  itself, since a chain of length  $\delta$  with  $|\delta| \leq \lambda$  of filters with bases of size at most  $\lambda$  has a union with a base of size at most  $\lambda \cdot \lambda = \lambda$ .

**Theorem 5.2** (Closure at the character bound [10 Thm 5.1]). *Let  $\lambda$  be an infinite cardinal, regular or singular. Every chain of successively stronger conditions in  $\mathbb{P}_{\lambda^+}$  of length less than  $\lambda^+$  has a common stronger condition. Consequently, any family of at most  $\lambda$  maximal antichains of  $\mathbb{C}_{\lambda^+}$  has a common refining maximal antichain.*

*Proof.* The first assertion is Proposition 5.1 at the regular cardinal  $\lambda^+$ . Now let  $(\mathcal{A}_\xi)_{\xi < \delta}$  be maximal antichains in  $\mathbb{C}_{\lambda^+}$ , where  $|\delta| \leq \lambda$ . Given a nonzero element  $w$ , first choose a basic element  $K_F \leq w$ . Recursively strengthen its filter to  $F_\xi$  so that  $K_{F_\xi}$  lies below a member of  $\mathcal{A}_\xi$ . Such a strengthening exists because the antichain has join 1 and the sets  $K_G$  are order dense. At limit stages and at the final stage, take the union of the earlier filters. The first assertion makes this a legal construction. The final  $K_G$  is below  $w$  and below one member of each  $\mathcal{A}_\xi$ .

Hence the nonzero elements refining all the antichains form an order-dense set. Choose a maximal pairwise disjoint subfamily of that set. It has join 1 by density and maximality, so it is the required common refinement.  $\square$

We call the conclusion *partition distributivity for at most  $\lambda$  families*. This explicit terminology avoids conventions about whether “ $\lambda$ -distributive” means a bound of  $< \lambda$  or a bound of  $\leq \lambda$ . Equivalently, an intersection of at most  $\lambda$  dense open subsets of the forcing poset  $\mathbb{P}_{\lambda^+}$  is dense. The proof just given is the usual closure-to-distributivity argument, specialized to filter bases.

**Corollary 5.3** (Local simultaneous decisions [10 Cor. 5.2]). *For every family  $(b_\xi)_{\xi < \delta}$  in  $\mathbb{C}_{\lambda^+}$  with  $|\delta| \leq \lambda$ , there is a maximal antichain  $\mathcal{A}$  such that each  $a \in \mathcal{A}$  lies below either  $b_\xi$  or  $\neg b_\xi$  for every  $\xi < \delta$ .*

*Proof.* Apply Theorem 5.2 to the nonzero members of the binary partitions  $\{b_\xi, \neg b_\xi\}$ .  $\square$

The antichain in this result is a local cover, not a claim that the original family has one global constant value. Different regions of the cover can give different sequences of decisions.

## 5.1 At which stages the closure consequences are proved

Source 10's results that go through Theorem 5.2 are Corollary 5.3, the bound  $\aleph_0 < \mathfrak{h}$  and Corollary 6.3 in Section 6, Proposition 8.1, Proposition 11.9, Theorems 13.1 and 13.2, and the remark that the stages are countably distributive. They are stated here, as there, at the successor stages  $\mathbb{P}_{\lambda^+}$ .

*Merge note.* The other stages of source 08 are covered as follows.

- *The finite-generation stage  $\mathbb{P}_\omega$ .* It is countably closed: a countable chain  $F_{A_0} \subseteq F_{A_1} \subseteq \dots$  is bounded by  $F_X$  for any infinite pseudointersection  $X$  of the  $A_n$  (Lemma 4.1). Statements about the complete algebra  $\mathbb{C}_{\aleph_1}$  and its mixing model apply to  $\mathbb{C}_\omega$  through the marked isomorphism of Corollary 4.6 and its transport of mixing models (Corollary 12.5).
- *Regular limit stages.* Proposition 5.1 makes  $\mathbb{P}_\kappa < \kappa$ -closed. Neither source derives source 10's consequences there, and this report does not claim them.
- *Singular stages.* At a singular  $\kappa$  of countable cofinality, for example  $\mathbb{P}_{\aleph_\omega}$ , an increasing chain  $(F_n)_{n < \omega}$  with  $\chi_*(F_n) = \aleph_n$  can have a union needing  $\aleph_\omega$  generators, which lies outside the site. The union argument then gives no lower bound, and none of the results listed above is proved there. Whether such a stage is even countably closed is not decided here; it is part of question Q5.

The comparison theorems of Section 4, including the reverse barrier, the atom and point results of Section 7, and the exact sizes of Section 15 use no closure and hold at every stage for which they are stated.

## 6 The distributivity number as a local-reading boundary

This section, from source 10, identifies the first obstruction to simultaneous binary decisions in the countably generated completion  $\mathbb{C}_\omega (= \mathbb{C}_{\aleph_1})$ . The cardinal being identified is classical; the interpretation concerns the completed sections constructed in Section 11.

**Definition 6.1** ([10 Def. 6.1]). Let  $\mathfrak{h}$  be the least cardinality of a family of dense open subsets of  $\mathbb{B}^+$  whose intersection is not dense. Here a subset of  $\mathbb{B}^+$  is open when it is closed under passing to smaller nonzero Boolean elements. A family  $(b_\xi)_{\xi < \mu}$  of elements of  $\mathbb{C}_\omega$  is *locally jointly decidable* if

$$\{a \in \mathbb{B}^+ : \text{for every } \xi < \mu, e_\omega(a) \leq b_\xi \text{ or } e_\omega(a) \leq \neg b_\xi\} \quad (14)$$

is dense in  $\mathbb{B}^+$ .

This form of  $\mathfrak{h}$  is the usual distributivity number for  $\mathcal{P}(\omega)/\text{Fin}$  [3]. One has  $\aleph_0 < \mathfrak{h} \leq \mathfrak{c}$ . The first inequality follows from the countable case of Theorem 5.2 (at  $\lambda = \aleph_0$ ) and (13). For the upper bound, take the  $\mathfrak{c}$  many dense sets that decide each element of  $\mathbb{B}$ . No nonzero element of the atomless algebra  $\mathbb{B}$  can decide all its elements: splitting a representative infinite set into two infinite pieces disproves that possibility.

**Theorem 6.2** (Exact binary local-reading boundary [10 Thm 6.2]). *The least size of a family in  $\mathbb{C}_\omega$  that is not locally jointly decidable is  $\mathfrak{h}$ .*

*Proof.* For  $b \in \mathbb{C}_\omega$ , the set

$$\mathcal{D}_b = \{a \in \mathbb{B}^+ : e_\omega(a) \leq b \text{ or } e_\omega(a) \leq \neg b\}$$

is dense open. Indeed, at least one of  $e_\omega(a) \wedge b$  and  $e_\omega(a) \wedge \neg b$  is nonzero, and order density supplies a nonzero old event below that part. Thus fewer than  $\mathfrak{h}$  binary events have a dense set of joint decisions by the definition of  $\mathfrak{h}$ .

For the converse, choose dense open sets  $(\mathcal{D}_\xi)_{\xi < \mathfrak{h}}$  with nondense intersection. For each  $\xi$ , choose a maximal antichain  $\mathcal{A}_\xi \subseteq \mathcal{D}_\xi$  in  $\mathbb{B}^+$ . It has join 1 in  $\mathbb{C}_\omega$ . Since  $|\mathbb{B}| = \mathfrak{c}$ , its members admit pairwise distinct binary real codes

$$\gamma_\xi : \mathcal{A}_\xi \longrightarrow 2^\omega.$$

For  $n < \omega$ , form the completed binary event

$$b_{\xi,n} = \bigvee \{e_\omega(a) : a \in \mathcal{A}_\xi, \gamma_\xi(a)(n) = 1\}. \quad (15)$$

There are at most  $\mathfrak{h} \cdot \aleph_0 = \mathfrak{h}$  such events.

Suppose  $a \in \mathbb{B}^+$  decides every event in (15). Fix  $\xi$ . The region  $e_\omega(a)$  can meet at most one member of  $e_\omega(\mathcal{A}_\xi)$ . If it met two, some bit of their distinct codes would differ, so  $e_\omega(a)$  would meet both  $b_{\xi,n}$  and its complement, contrary to the decision of that bit. It meets at least one member, since the antichain has join 1. Finite-meet distribution over arbitrary joins now implies that  $e_\omega(a)$  lies below that unique member. Therefore  $a \in \mathcal{D}_\xi$  because  $\mathcal{D}_\xi$  is downward closed. This holds for every  $\xi$ , so every joint decision lies in  $\bigcap_{\xi < \mathfrak{h}} \mathcal{D}_\xi$ , which is not dense. The family (15) is the desired obstruction.  $\square$

The coding by reals is important. A dense-set witness initially gives partitions with potentially many cells, not binary partitions. The countably many bits of the code reduce each such partition to binary events without changing the infinite cardinal  $\mathfrak{h}$ .

**Corollary 6.3** (An unlabeled nonisomorphism [10 Cor. 6.3]). *Let  $\kappa = \lambda^+$  be a successor cardinal with  $\lambda \geq \mathfrak{h}$  (equivalently, a successor cardinal  $\kappa > \mathfrak{h}$ ). Then  $\mathbb{C}_\kappa$  is not isomorphic to  $\mathbb{C}_\omega$ , even by an isomorphism that need not preserve the old events. The arguments also recover the classical inequality  $\mathfrak{p} \leq \mathfrak{h}$ .*

*Proof.* By Theorem 5.2, every  $\mathfrak{h}$  many binary partitions in  $\mathbb{C}_{\lambda^+}$  have a common maximal-antichain refinement, since  $\mathfrak{h} \leq \lambda$ . By Theorem 6.2, this property fails in  $\mathbb{C}_\omega$ : such a refinement would, using the order-dense copy of  $\mathbb{B}$ , make the decision set (14) dense. The property is invariant under Boolean isomorphism. Finally, if  $\mathfrak{h} < \mathfrak{p}$ , the successor stage  $\kappa = \mathfrak{h}^+$  satisfies  $\kappa \leq \mathfrak{p}$ , so Corollary 4.5(a) gives  $\mathbb{C}_{\mathfrak{h}^+} \cong \mathbb{C}_\omega$ , contradicting the first assertion.  $\square$

We emphasize that recovering  $\mathfrak{p} \leq \mathfrak{h}$  is a consistency check on the framework, not a claim of a new cardinal-characteristic inequality. For a possible successor stage  $\mathfrak{p} < \kappa \leq \mathfrak{h}$ , the two-sided *labeled* barrier of Corollary 4.5 still holds, while this particular unlabeled argument does not decide isomorphism. Corollary 6.3 is proved only at successor stages, as explained in Section 5.1.

*Merge note.* Corollary 6.3 partly answers source 08's question Q4 (marked versus unmarked isomorphism): at the successor stages above  $\mathfrak{h}$  the unmarked question has a negative answer. Q4 is re-scoped accordingly in Section 19.

## 7 The ultrafilter threshold: atoms, points, and atomicity

### 7.1 The exact atom spectrum

Define

$$S_\kappa = \{U \in \omega^* : \chi_*(U) < \kappa\}.$$

This is a set of ultrafilters, not a set of hypothetical generic filters in an extension. Source 10's  $\mathcal{U}_\lambda$ , the free ultrafilters of base character at most  $\lambda$ , is  $S_{\lambda^+}$  by Lemma 2.2. The topology  $\tau_\kappa$  is Hausdorff because it contains the original Stone topology.

**Theorem 7.1** (Atoms are exactly small-character ultrafilters [08 Thm 5.1; 10 Thm 7.3, Cor. 7.4]). *The atoms of  $\mathbb{C}_\kappa$  are precisely the singleton sets  $\{U\}$  with  $U \in S_\kappa$ . Consequently*

$$\mathbb{C}_\kappa \text{ is atomless} \iff \kappa \leq \mathfrak{u}.$$

*The first cardinal stage with atoms is  $\mathfrak{u}^+$ .*

*Proof.* A nonempty regular open set containing two distinct points can be split by an original Stone clopen separating those points. Its intersections with that clopen and its complement are nonzero disjoint regular open sets. Such a set cannot be an atom. Every atom is therefore a singleton.

A singleton  $\{U\}$  is open in  $\tau_\kappa$  exactly when a basic  $K_F$  contains  $U$  and is contained in  $\{U\}$ . In that case  $K_F = K_U$ , and Lemma 3.2 gives  $F = U$ ; thus  $\chi_*(U) < \kappa$ . Conversely, if  $U \in S_\kappa$ , its singleton is the basic clopen  $K_U$ , and hence is an atom. By the definition of  $\mathfrak{u}$ , the set  $S_\kappa$  is empty exactly for  $\kappa \leq \mathfrak{u}$ .  $\square$

*Second proof, on the poset side* [source 10]. A singleton ultrafilter cone  $\uparrow U = \{U\}$  is a nonzero regular open with no nonzero proper part, so it is an atom of  $\text{RO}(\mathbb{P}_\kappa)$ . Conversely, any atom contains a cone  $\uparrow F$ , and must equal that cone. If  $F$  is not an ultrafilter, choose  $A$  with neither  $A$  nor  $\omega \setminus A$  in  $F$ . Adjoining these two sets separately produces two proper filters in  $\mathbb{P}_\kappa$  (Lemma 2.4), with disjoint nonzero cones below  $\uparrow F$ . This contradicts atomicity. Therefore  $F$  is an ultrafilter of  $\mathbb{P}_\kappa$ , and Lemma 3.6 transfers the conclusion to  $\mathbb{C}_\kappa$ .  $\square$

**Proposition 7.2** (Geometric points of the Booleanization [08 Prop. 5.2; 10 Lemma 7.5, Cor. 7.6]). *The points of the localic topos  $\text{Sh}(\mathbb{P}_\kappa, J_{\text{dense}})$ , equivalently the complete Boolean points of  $\mathbb{C}_\kappa$  (unital maps  $\mathbb{C}_\kappa \rightarrow 2$  preserving finite meets and arbitrary joins), are indexed by  $S_\kappa$ . In particular, there are none when  $\kappa \leq \mathfrak{u}$ .*

*Proof.* A point of a locale is a map from its frame to 2 preserving finite meets and arbitrary joins; such a map preserves complements automatically. Let  $q : \mathbb{E} \rightarrow 2$  be such a map from a nontrivial complete Boolean algebra  $\mathbb{E}$ . Put

$$z = \bigvee \{b \in \mathbb{E} : q(b) = 0\}, \quad a = \neg z.$$

Join preservation gives  $q(z) = 0$ , so  $q(a) = 1$  and  $a \neq 0$ . If  $0 < b \leq a$ , then  $q(b)$  cannot be zero, because that would imply  $b \leq z$ . If both  $b$  and  $a \wedge \neg b$  were nonzero, they would both map to 1, contradicting preservation of their zero meet. Thus  $a$  is an atom. Conversely an atom  $a$  defines  $q_a(b) = 1$  exactly when  $a \leq b$ ; an atom lies below an arbitrary join exactly when it lies below one of the terms (distribute its meet over the join), so  $q_a$  preserves arbitrary joins. These constructions are inverse. Apply Theorems 3.3 and 7.1.  $\square$

The two sources give this same kernel-join proof.

**Corollary 7.3** ([10 Cor. 7.6]). *There is no unital complete Boolean homomorphism from  $\mathbb{C}_\omega$  into any nonzero power-set algebra.*

*Proof.* The completion  $\mathbb{C}_\omega$  is atomless, because below each nonzero element there is an old infinite-set event that can be split into two nonzero events. A complete map  $\mathbb{C}_\omega \rightarrow \mathcal{P}(I)$  with  $I \neq \emptyset$ , followed by evaluation at any  $i \in I$ , would be a complete Boolean point of  $\mathbb{C}_\omega$ .  $\square$

At the full-site endpoint  $\mathbb{C}_{\epsilon^+} = \mathcal{P}(\omega^*)$  this is stronger than the event-preserving obstruction in the direction from  $\mathbb{C}_\omega$ . In the other direction, complete maps  $\mathcal{P}(\omega^*) \rightarrow \mathbb{C}_\omega$  certainly exist abstractly: evaluation at a fixed ultrafilter, followed by the inclusion  $2 \hookrightarrow \mathbb{C}_\omega$ , is one. What Theorem 4.4 rules out is preservation of *all* the original events by such a map.

*Remark 7.4* (Do not confuse two kinds of points [08 §5.1; 10 Remark 7.7]). This does not deny that  $X = \omega^*$  contains many ordinary points. An atomless complete Boolean algebra still has ordinary ultrafilters in ZFC. Those are points of its *Stone space* and yield maps to 2 preserving finite Boolean operations; they generally do not preserve arbitrary joins. Booleanization can therefore have no geometric points even though the topological presentation from which it was obtained has points: a point-free Boolean locale need not have an empty Stone space. This distinction is essential in interpreting the countable-site model. The model-theoretic specializations of Section 12 use ordinary ultrafilters, not complete points.

## 7.2 Full atomicity is an extension property

A complete Boolean algebra is atomic if every nonzero element contains an atom. Having one atom, or many atoms, is weaker.

**Theorem 7.5** (Atomicity and the canonical decomposition [08 Thm 5.3; 10 Thm 7.3]). *For every infinite  $\kappa$ , the following are equivalent:*

- (i)  $\mathbb{C}_\kappa$  is atomic;
- (ii)  $S_\kappa$  is dense in  $(X, \tau_\kappa)$ ;
- (iii) every  $F \in \mathbb{P}_\kappa$  extends to an ultrafilter  $U \in \mathbb{P}_\kappa$ .

Let

$$a_\kappa = \bigvee_{U \in S_\kappa} \{U\} = \text{int}_{\tau_\kappa} \text{cl}_{\tau_\kappa} S_\kappa, \quad r_\kappa = \neg a_\kappa.$$

There is a canonical complete Boolean decomposition

$$\mathbb{C}_\kappa \cong \mathcal{P}(S_\kappa) \times (\mathbb{C}_\kappa \upharpoonright r_\kappa), \quad (16)$$

and the second factor is atomless. It is the trivial one-element algebra exactly when the equivalent conditions hold. On the poset side (Lemma 3.6) the residual element is

$$r_\kappa = \{F \in \mathbb{P}_\kappa : \text{no } U \in S_\kappa \text{ extends } F\}. \quad (17)$$

*Proof.* Every nonempty regular open contains an atom exactly when it meets  $S_\kappa$ , by Theorem 7.1. This is equivalent to density. Since the  $K_F$  form a base, density is equivalent to  $K_F \cap S_\kappa \neq \emptyset$  for every allowed  $F$ , which is precisely (iii).

For any complete Boolean algebra and any element  $a$ , the map  $b \mapsto (b \wedge a, b \wedge \neg a)$  gives the product of its two relative algebras. Below the join of all atoms, an element is uniquely the join of the atoms below it. Thus the first relative algebra is the power set of its atoms, here  $S_\kappa$ . The complementary relative algebra contains no atom: an atom in it would be an atom of the original algebra and hence lie below  $a_\kappa$ , a contradiction. Taking the negation (5) of the union of the singleton cones  $\uparrow U$ ,  $U \in S_\kappa$ , gives (17).  $\square$

Source 10 states (17) at its stages; the proof uses no closure and holds at every stage. In source 10's words,  $\mathbb{C}_{\lambda^+}$  is wholly atomic if and only if every filter of character at most  $\lambda$  extends to an ultrafilter of character at most  $\lambda$ .

**Corollary 7.6** (The full site [10 Cor. 7.4]). *For  $\kappa > \mathfrak{c}$  the algebra  $\mathbb{C}_\kappa = \mathbb{C}_{\mathfrak{c}^+}$  is wholly atomic and*

$$\mathbb{C}_{\mathfrak{c}^+} = \mathcal{P}(\omega^*), \quad e_{\mathfrak{c}^+}([A]) = \{U \in \omega^* : A \in U\}. \quad (18)$$

*Proof.* At the all-filter stage every proper filter extends to an ultrafilter, and that ultrafilter has at most  $c$  generators. Thus the residual factor is zero. The event formula follows by evaluating membership at each atom; it is also (4).  $\square$

The endpoint (18) is classical in this setting; Massas identifies it as the canonical extension of the modulo-finite algebra under the ultrafilter lemma [2, Section 6.1]. “Canonical extension” there is not the same construction as the order-dense Boolean completion  $\mathbb{C}_\omega$ . Corollary 4.5 and Theorem 7.1 describe precisely why the two endpoints must be kept distinct.

Millán’s notation  $\text{GE}(\mathcal{U}, \kappa)$  expresses that every filter with a base of size below  $\kappa$  extends a member of a specified ultrafilter class  $\mathcal{U}$  [5, Definition 2.1]. For uncountable  $\kappa$ , the criterion above is exactly

$$\text{GE}(S_\kappa, \kappa).$$

This identifies full atomicity with a familiar but stronger extension problem, rather than silently deducing it from  $\mathfrak{u} < \kappa$ .

*Remark 7.7* (Why ordinary density is insufficient [08 Remark 5.4]). If  $S_\kappa$  is nonempty, it is dense in the *original* Stone topology. To see this, transport one small-character free ultrafilter along a bijection from  $\omega$  to any prescribed infinite  $A \subseteq \omega$ . The resulting ultrafilter contains  $A$  and has the same character. But ordinary Stone density says only that the sets  $K_{F_A}$  meet  $S_\kappa$ ; Theorem 7.5 requires this for every small-generated filter. We make no assertion that existence alone implies, or refutes, this stronger density. The least bound giving a wholly atomic algebra need not be determined by the point-existence statement of Theorem 7.1.

## 8 A forcing interpretation of the residual part

The decomposition (16) also has a direct set-theoretic meaning, found in source 10. We state it in the usual language of forcing over a ground model  $\mathbf{V}$  of ZFC. All cardinals and all posets in the statement are initially computed in  $\mathbf{V}$ . A generic extension is understood externally; no generic filter over the whole universe is being postulated inside that universe.

**Proposition 8.1** (Old and new ultrafilter regions [10 Prop. 8.1]). *Let  $\lambda$  be an infinite cardinal. Forcing with  $\mathbb{P}_{\lambda^+}$  adds no new sequences of ground-model elements of length at most  $\lambda$ . If  $G$  is generic, then*

$$U_G = \bigcup_{F \in G} F$$

*is a free ultrafilter on  $\omega$  in the extension. On the atomic part  $a_{\lambda^+}$  it is a ground-model ultrafilter in  $S_{\lambda^+}$ , that is, of character at most  $\lambda$ . On a nonzero residual part  $r_{\lambda^+}$ , it is a new ultrafilter and has character greater than  $\lambda$  in the extension.*

*Proof.* For a name for a sequence of length at most  $\lambda$ , recursively strengthen a condition to decide each entry and take unions at limit stages and at the end. Theorem 5.2 supplies all lower bounds. The final condition decides the whole sequence as a sequence in  $\mathbf{V}$ . This standard decision argument proves the first assertion. In particular there are no new subsets of  $\omega$ .

The generic family is directed in the strengthening order, so its union is a proper free filter. For every ground-model  $A \subseteq \omega$ , the set of filters deciding  $A$  is dense by Lemma 2.4. Thus the union decides every ground-model subset, and, since there are no new subsets of  $\omega$ , it is an ultrafilter in the extension. If a generic condition is a singleton atom  $U$ , its stronger cone consists only of  $U$ , and the union is  $U$ .

Suppose now that we force below the residual region. No ground-model ultrafilter in  $S_{\lambda^+}$  can be the generic union, by (17). For a fixed ground-model ultrafilter  $W$  with  $\chi^{\mathbf{V}}(W) > \lambda$ , the conditions  $F$  not contained in  $W$  form a dense set: if  $F \subseteq W$ , choose  $A \in W \setminus F$  and adjoin  $\omega \setminus A$  to  $F$ . The resulting proper extension is not contained in  $W$ . The generic union therefore differs from every ground-model ultrafilter.

If  $U_G$  had a base of size at most  $\lambda$  in the extension, enumerate that base in a sequence of length at most  $\lambda$ , with repetition if necessary. Every entry is a ground-model subset of  $\omega$ , and the first paragraph says the entire sequence belongs to  $\mathbf{V}$ . Its generated filter would then be a ground-model ultrafilter, since the extension has no new subsets of  $\omega$ . This contradicts newness. Thus its character in the extension exceeds  $\lambda$ .  $\square$

At the countably generated stage this describes a familiar modulo-finite forcing behavior: new ultrafilters can be added without new reals. At the full-filter stage the forcing is atomic and merely selects an already available ultrafilter. The bounded-character residual interpolates between these behaviors, without asserting that its intermediate stages are all isomorphic, and without identifying the residual forcing uniquely. The proposition uses Theorem 5.2 and is therefore stated only at the successor stages (Section 5.1).

## 9 Early atomicity and genuinely different atomic stages

The full stage  $\mathbb{C}_{\mathfrak{c}^+} = \mathcal{P}(\omega^*)$  is atomic. Is full filter complexity necessary for atomicity? The answer is no, relative to a published forcing model. The following is the only imported relative-consistency input in this report.

**Proposition 9.1** (Published model input; **imported** [08 Prop. 6.1]). *Millán's proof of Theorem 3.14 [5, pp. 34–35] supplies, relative to consistency of ZFC, a model with*

$$\mathfrak{c} = 2^{\aleph_1} = \aleph_2, \quad \text{GE}(\mathcal{Pt}, \mathfrak{c}), \quad |\mathcal{Pt}| = \mathfrak{c},$$

where  $\mathcal{Pt}$  is the class of  $P$ -point ultrafilters, and every member of  $\mathcal{Pt}$  has character  $\aleph_1$ .

Here a  $P$ -point is a free ultrafilter in which every countable family has a pseudointersection belonging to the ultrafilter. We only use the displayed properties, not additional special features of  $P$ -points. The cited construction is a countable-support iteration of rational perfect set forcing. We do not claim to have proved that forcing theorem; Proposition 9.1 is an imported input, not a result of this report.

**Lemma 9.2** (The number of free ultrafilters [08 Lemma 6.2; 10 Lemma 12.3]). *In ZFC,  $|\omega^*| = 2^{\mathfrak{c}}$ .*

*Proof.* The upper bound follows because an ultrafilter is a subset of  $\mathcal{P}(\omega)$ . For the lower bound, use the countably infinite set

$$I = \{(n, H) : n < \omega, H \subseteq 2^n\}.$$

It is countably infinite because each of its levels is finite. For each real  $x \in 2^\omega$ , put  $A_x = \{(n, H) : x \upharpoonright n \in H\}$ . Given finitely many distinct reals, their initial segments are distinct for all sufficiently large  $n$ . At every such  $n$  choose  $H$  to include exactly the initial segments prescribed by an arbitrary binary pattern. Thus every finite Boolean combination of the  $A_x$  with no repeated index is infinite.

For every function  $\varepsilon : 2^\omega \rightarrow 2$ , the family choosing  $A_x$  when  $\varepsilon(x) = 1$  and its complement otherwise has SFIP. Together with the cofinite filter it extends to a free ultrafilter on  $I$ . Different functions force different ultrafilters, since they make opposite decisions about some  $A_x$ . Choice allows one extension to be selected for each function, giving  $2^{\mathfrak{c}}$  distinct ultrafilters. Transfer them to  $\omega$  by a bijection.  $\square$

The two sources give the same independent-family construction (source 10 indexes it by  $\Omega$  and writes  $I_x$  for  $A_x$ ).

**Theorem 9.3** (A three-stage hierarchy from the published model [08 Thm 6.3]). *In the model of Proposition 9.1,*

|                                           | <i>stage</i>        | <i>Booleanization</i>                                               |
|-------------------------------------------|---------------------|---------------------------------------------------------------------|
| $\mathfrak{p} = \mathfrak{u} = \aleph_1,$ | $\kappa = \aleph_1$ | $\mathbb{C}_{\aleph_1} \cong \text{RO}(\omega^*), \text{ atomless}$ |
|                                           | $\kappa = \aleph_2$ | $\mathbb{C}_{\aleph_2} \cong \mathcal{P}(\mathfrak{c})$             |
|                                           | $\kappa = \aleph_3$ | $\mathbb{C}_{\aleph_3} \cong \mathcal{P}(2^{\mathfrak{c}}).$        |

*In particular, the last two complete Boolean algebras are atomic but are not isomorphic, even without marking the original events.*

*Proof.* The existence of ultrafilters with character  $\aleph_1$  gives  $\mathfrak{u} = \aleph_1$ , and Lemma 4.1 then gives  $\mathfrak{p} = \aleph_1$ . The first line follows from Corollary 4.6.

Every  $F \in \mathbb{P}_{\aleph_2}$  has a  $P$ -point extension, and every such extension has character  $\aleph_1$ . Thus the small-character ultrafilters are dense in this site, and Theorem 7.5 makes  $\mathbb{C}_{\aleph_2}$  atomic. There are at least  $\mathfrak{c}$  such ultrafilters, by the stated number of  $P$ -points. There are at most  $\mathfrak{c}^{\aleph_1} = \mathfrak{c}$  generating families of size at most  $\aleph_1$ : use  $\mathfrak{c} = 2^{\aleph_1}$  to compute  $(2^{\aleph_1})^{\aleph_1} = 2^{\aleph_1}$ . Every member of  $\mathcal{S}_{\aleph_2}$  is determined by one such family. Therefore  $|\mathcal{S}_{\aleph_2}| = \mathfrak{c}$ , proving the middle line.

Finally  $\aleph_3 = \mathfrak{c}^+$ , so the last line follows from (4) and Lemma 9.2. The two atomic algebras have respectively  $\mathfrak{c}$  and  $2^{\mathfrak{c}}$  atoms; Cantor’s theorem distinguishes their atom sets and hence the algebras.  $\square$

The conclusion settles the proposed “atomic only at the full site” possibility negatively, with its exact external assumption exposed. It does not settle the general atomicity criterion in terms of a short list of cardinal characteristics. Notice also that two atomic forcing notions can both add no new sets in the usual generic-extension sense and still have nonisomorphic complete Boolean algebras and non-equivalent localic sheaf categories. The atom counts here retain information that the phrase “trivial forcing” discards.

For comparison, under CH we have  $\mathfrak{p} = \mathfrak{u} = \mathfrak{c} = \aleph_1$ . Then the countable-base stage is atomless, while the very next cardinal stage is already the full atomic site. The model above separates atomicity from the admission of all filters.

*Merge note.* Theorem 9.3 answers the second clause of source 10’s research question 14.2 (question R2 here), “Can this extension property hold strictly below  $\mathfrak{c}$ ?”, positively, relative to the imported model: in source 10’s indexing the stage  $\mathcal{F}_{\aleph_1} = \mathbb{P}_{\aleph_2}$  is wholly atomic while  $\aleph_1 < \mathfrak{c} = \aleph_2$ . The answer is a relative-consistency result that rests on Proposition 9.1; it is not a ZFC theorem. R2 is re-scoped in Section 19.

## 10 A countable toy model and the marking issue

The following elementary example isolates the complete-homomorphism obstruction without requiring uncountable cardinal characteristics. It is not a claim of a new finite–cofinite algebra phenomenon. Pierobon and Viale also use that algebra to illustrate delicate completion behavior [4, Remark 3.7]; our purpose is to display the two filter stages explicitly.

Let  $\mathbb{B}_0$  be the Boolean algebra of finite and cofinite subsets of  $\omega$ . Its Stone space is  $Y = \omega \cup \{\infty\}$ , where each integer is isolated and neighborhoods of  $\infty$  are cofinite. For an abstract Boolean algebra, repeat the filter construction using proper filters generated by fewer than  $\kappa$  Boolean elements. At the finite-generation stage, the topology is the original Stone topology. Since  $\omega$  is a dense set of isolated points,

$$\text{RO}(Y) \cong \mathcal{P}(\omega).$$

The embedding of  $\mathbb{B}_0$  is the ordinary inclusion into  $\mathcal{P}(\omega)$ . The isomorphism sends a regular open set to its intersection with  $\omega$ ; its inverse is regularization of that subset in  $Y$ .

At the countable-generation stage, all proper filters of  $\mathbb{B}_0$  are allowed. In particular, the cofinite ultrafilter corresponds to  $\{\infty\}$ , which becomes open. The topology is now discrete, and the completion is  $\mathcal{P}(\omega \cup \{\infty\})$ . A finite Boolean element  $A$  maps to  $A$ , while a cofinite one maps to  $A \cup \{\infty\}$ .

**Proposition 10.1** (Abstractly isomorphic, but not compatibly complete [08 Prop. 7.1]). *The two complete Boolean algebras just described are abstractly isomorphic. No complete Boolean homomorphism from the first to the second extends the identity of  $\mathbb{B}_0$ .*

*Proof.* Their atom sets are both countably infinite, so a bijection of those sets induces an abstract complete isomorphism. On the other hand, for  $T_n = \{m : m \geq n\}$ , the meet of the embedded  $T_n$  is empty in the first algebra and  $\{\infty\}$  in the second. A complete homomorphism extending the original embedding would preserve that meet, which is impossible. Equivalently, the join of the singletons  $\{n\}$  is 1 in the first algebra but not in the second.  $\square$

Thus “the completions differ over the original events” is a genuine mathematical statement, but it must not be strengthened without proof to “the completions are abstractly nonisomorphic.” The example supplies a concrete audit test for the wording of Theorem 4.3.

## 11 Reduced powers and their explicit mixing sheaves

### 11.1 The raw presheaf is separated

Let  $M$  be a nonempty set. For a free filter  $F$ , define

$$\mathcal{R}_M(F) = M^\omega / \sim_F, \quad f \sim_F g \iff E(f, g) := \{n : f(n) = g(n)\} \in F.$$

For  $G \supseteq F$ , restriction sends  $[f]_F$  to  $[f]_G$ . These definitions make a presheaf on every  $\mathbb{P}_\kappa$ . Its global sections at  $\text{Fr}$  are the raw sequences modulo eventual equality.

**Proposition 11.1** ([08 Prop. 8.1; in 10 inside Thm 9.2]). *The presheaf  $\mathcal{R}_M$  is separated for the relative dense topology on  $\mathbb{P}_\kappa$ . Its map into its associated sheaf is injective at every object.*

*Proof.* If  $E(f, g) \notin F$ , adjoining its complement gives an allowed proper extension  $H$  of  $F$ . At every further extension of  $H$ , the two sequences remain inequivalent. A dense cover of  $F$  must reach some such extension. Hence two inequivalent sections cannot become equal throughout a dense cover. The same fact gives injectivity of the sheafification unit; it will also follow directly from the construction below.  $\square$

Separatedness is not sheafness. It says that gluing, when it exists, is unique; it does not supply a representative for every matching family. On the full site separatedness is the earlier report’s Theorem 5.1 [1].

### 11.2 Mixtures over a Boolean element

Fix  $\kappa$ , abbreviate  $e = e_\kappa$ , and write  $\sigma_F = K_F$ . An antichain below  $b \in \mathbb{C}_\kappa$  is a set of pairwise disjoint nonzero elements, each at most  $b$ . It is a partition of  $b$  when its join is  $b$ .

**Definition 11.2** (A mixture [08 Def. 8.2; 10 §9.1]). A mixture over  $b$  is a partition  $P \subseteq \mathbb{C}_\kappa^+$  of  $b$ , together with one sequence  $f_p \in M^\omega$  for each  $p \in P$ . We write it as  $x = \sum_{p \in P} p \cdot f_p$ ; the notation denotes pasting, not numerical addition. Over 0 the empty partition is the unique representation.

If  $x = \sum p \cdot f_p$  and  $y = \sum q \cdot g_q$  are mixtures on the same support  $b$ , define their equality value by

$$\|x = y\| = \bigvee_{p,q} p \wedge q \wedge e([E(f_p, g_q)]). \quad (19)$$

Identify  $x$  and  $y$  when this value is  $b$ . Let  $\text{Mix}_M(b)$  be the resulting set of equivalence classes.

All representations form a set: partitions are subsets of the set  $\mathbb{C}_\kappa$ , and labels are chosen from the set  $M^\omega$ . There is no proper-class collection of names in this construction. The relevant labels are *sequences in  $M$* , not just constant members of  $M$ ; that distinction is essential for infinite  $M$  (Example 11.8).

**Lemma 11.3** (Equality, refinement, restriction, and flattening [08 Lemma 8.3; 10 Lemma 9.1]). *Equality at support  $b$  is an equivalence relation. Replacing a partition by a refinement and retaining its labels does not change the mixture. For  $c \leq b$ , intersecting each partition element with  $c$  and dropping zeros defines a well-defined restriction  $\text{Mix}_M(b) \rightarrow \text{Mix}_M(c)$ . A mixture of mixtures along a partition of  $b$  is represented by a single mixture over  $b$ .*

*Proof.* Reflexivity follows because the diagonal terms of (19) join to  $b$ ; symmetry is immediate. For transitivity, refine three partitions simultaneously. On a cell  $p \wedge q \wedge r$ , the ordinary implication

$$E(f_p, g_q) \cap E(g_q, h_r) \subseteq E(f_p, h_r)$$

becomes a Boolean inequality after applying  $e$ . Taking joins over all cells gives  $\|x = y\| \wedge \|y = z\| \leq \|x = z\|$ . Thus equality at  $b$  is transitive. The same formula shows that refinement preserves equality values. Intersecting every term with  $c$  multiplies the equality value by  $c$ , proving that restriction is independent of representatives. Successive restrictions agree by associativity of Boolean meet.

For flattening, if the outer cell is  $b_i$  and the inner mixture uses cells  $c_{i,j}$  with labels  $f_{i,j}$ , use all nonzero cells  $b_i \wedge c_{i,j}$  with labels  $f_{i,j}$ . They are pairwise disjoint and have join  $b$ , and the equality value with each intended inner element is at least its outer cell.  $\square$

The use of arbitrary joins here is legitimate in a complete Boolean algebra. Finite meet distributes over arbitrary joins: for fixed  $a$ , the operation  $x \mapsto a \wedge x$  is left adjoint to  $x \mapsto \neg a \vee x$ . No stronger complete-distributivity assumption is being made.

**Lemma 11.4** (The mixing sheaf [08 Lemma 8.4; 10 Thm 9.2, first step]). *The functor  $b \mapsto \text{Mix}_M(b)$  is a sheaf on the complete Boolean algebra  $\mathbb{C}_\kappa$  with its join coverage.*

*Proof.* Suppose  $b = \bigvee_i b_i$  and  $x_i \in \text{Mix}_M(b_i)$  agree on all overlaps. Choose a maximal disjoint family of nonzero elements each below some  $b_i$ . Its join is  $b$ : otherwise its nonzero complement inside  $b$  would meet some  $b_i$  and permit one more element. For each cell choose such an  $i$ , restrict  $x_i$  to the cell, and flatten the resulting partitions (Lemma 11.3). Their union is a partition of  $b$  labelled by sequences, hence a mixture  $x$ .

On overlap with any  $b_j$ , each of these pieces agrees with  $x_j$  by compatibility of the original family. Formula (19) therefore gives equality value  $b_j$ , so  $x$  is an amalgam. If two amalgams exist, their equality value contains every  $b_i$  and hence their join  $b$ . They represent the same element.  $\square$

**Remark 11.5** (Source 10's presentation of the same object). Source 10 uses only *global* presentations  $\sum_i b_i \cdot f_i$  over partitions of 1, calls the quotient by value-1 equality  $S_\lambda(A)$ , and defines the sections on a region  $w$  as the quotient  $S_\lambda(A; w)$  by the relation  $w \leq \|s = t\|$ . A presentation on a partition of  $w$  extends globally by an arbitrary fixed constant label on  $\neg w$ , which is why global presentations suffice; at  $w = 0$  there is one section. Restricting a global presentation to  $w$  identifies  $S_\lambda(A; w)$  with  $\text{Mix}_M(w)$  at stage  $\lambda^+$  (with  $A = M$ ). Its proof

of the associated-sheaf theorem follows the same steps as the one below (disjoint refinement and flattening, the raw equality test, local representability by ordinary sequences, and the universal gluing argument), so it is not printed as a separate route. The general theory of associated sheaves is standard [7]; the point of both sources is that here it can be made explicit.

### 11.3 The associated-sheaf theorem

**Theorem 11.6** (Explicit sheafification on every complexity stage [08 Thm 8.5; 10 Thm 9.2]). *The associated sheaf of  $\mathcal{R}_M$  on  $\mathbb{P}_\kappa$  is*

$$\mathcal{S}_{M,\kappa}(F) = \text{Mix}_M(\sigma_F). \quad (20)$$

*The unit sends  $[f]_F$  to the single-label mixture  $\sigma_F \cdot f$ . It is injective. Every completed section is locally represented by ordinary sequences. Global sections, meaning sections at the weakest filter  $\text{Fr}$ , are exactly  $\text{Mix}_M(1)$ .*

*Proof.* For two sequences on support  $\sigma_F$ , their equality value is  $\sigma_F \wedge e([E(f, g)])$ . By Lemma 3.2, this equals  $\sigma_F$  exactly when  $E(f, g) \in F$ . Thus the unit is well-defined and injective; it commutes with restrictions.

The functor in (20) is a sheaf by Lemma 11.4 and Theorem 3.3. It remains to verify its universal property, rather than merely exhibiting a larger sheaf. Let  $x$  be a mixture on  $\sigma_F$ . For every nonzero partition cell  $p$  of  $x$ , take all conditions  $\sigma_G \leq p$ . These conditions together cover  $\sigma_F$  by order density and the fact that the cells join to  $\sigma_F$ . On each such  $G$ , the mixture is represented by the raw sequence labelling  $p$ . Thus the unit is locally surjective.

Now let  $\alpha : \mathcal{R}_M \rightarrow T$  be a morphism into a sheaf  $T$ . On the covering conditions just constructed, apply  $\alpha$  to the local raw representatives of  $x$ . They agree in  $T$  on all overlaps: their equality as mixtures on any overlap is witnessed locally by conditions contained in their sequence-equality event, and Lemma 3.2 says that on these conditions their raw reduced-power classes are equal. Separatedness of  $T$  gives equality on the whole overlap. Glue in  $T$  to define the image of  $x$ .

Refining the local representations does not change the glued section, by uniqueness. The same uniqueness makes the construction natural in  $F$  and independent of the mixture chosen to represent  $x$ . It extends  $\alpha$ , and any extension must have these local values, so it is unique. This is the sheafification universal property. Finally  $\sigma_{\text{Fr}} = X = 1$ , giving the description of globals.  $\square$

This is a specialized mixification construction, not a new general sheafification method. The prior framework in [4] clarifies why fullness of a Boolean-valued model must not be confused with its having all mixtures. The explicit form above makes that distinction usable for the repository's changing filter sites.

**Corollary 11.7** (The binary and finite-valued cases [08 Cor. 8.6; 10 Prop. 9.3]). *For  $M = 2$ , there is a canonical Boolean-algebra identification*

$$\mathcal{S}_{2,\kappa}(F) \cong \mathbb{C}_\kappa \upharpoonright \sigma_F, \quad \text{in particular} \quad \text{Mix}_2(1) \cong \mathbb{C}_\kappa.$$

*For a finite nonempty set  $M$ , sections at  $F$  correspond exactly to  $M$ -indexed Boolean partitions of  $\sigma_F$ , with zero parts allowed.*

*Proof.* On a raw sequence  $f : \omega \rightarrow 2$ , the event for value 1 is  $e([f^{-1}(1)])$ . On a mixture it is the join of the corresponding piecewise events. This determines the mixture: on that event it is the constant 1, and on its complement in  $\sigma_F$  it is 0. Conversely every  $c \leq \sigma_F$  is represented by the mixture  $c \cdot 1 + (\sigma_F \wedge \neg c) \cdot 0$ . For a finite set, the same argument splits each sequence according to its finitely many fibers  $f^{-1}\{m\}$  and then

joins the pieces carrying each value. Finite preservation by  $e$  makes the resulting events a partition of the support. Distinct constant labels have equality value zero, so the partition is recovered from the section  $s$  as  $d_m = \|s = \bar{m}\|$ , where  $\bar{m}$  is the constant sequence.  $\square$

In particular, the binary associated sheaf on the countably generated site has global Boolean algebra  $\text{RO}(\omega^*)$ , while the full-site binary associated sheaf has global Boolean algebra  $\mathcal{P}(\omega^*)$ . This computes the missing subsite object rather than merely showing that a proposed product formula cannot work.

Theorem 6.2 therefore says that  $\mathfrak{h}$  is the first size of a family of completed binary sections on the countably generated site which is not locally jointly constant. Theorem 5.2 says that at most  $\lambda$  many completed finite-valued sections on  $\mathbb{P}_{\lambda^+}$  can always be made jointly constant on a dense cover.

**Example 11.8** (Infinite labels cannot be discarded [10 Example 9.4]). For  $M = \omega$ , the raw sequence  $f(n) = n$  satisfies  $\|f = \bar{m}\| = 0$  for every standard  $m < \omega$ . It cannot be a mixture of constant labels: on every nonzero cell with constant label  $m$ , its equality with that constant would have to be nonzero. Consequently  $\text{Mix}_M(1)$  is not, in general, just a Boolean-valued copy of the constant set  $M$ . The raw sequences retain the nonstandard part of the construction.

**Proposition 11.9** (Local simultaneous raw representatives [10 Prop. 9.5]). *Let  $\lambda$  be infinite. For any family of at most  $\lambda$  elements of  $\text{Mix}_M(1)$  at stage  $\lambda^+$ , there is a Boolean partition of 1 in  $\mathbb{C}_{\lambda^+}$  on each cell of which all members of the family are represented simultaneously by ordinary sequences.*

*Proof.* Choose one presentation for each member and apply Theorem 5.2 to their at most  $\lambda$  many partitions. A cell in a common refinement selects a raw label for every member.  $\square$

For infinite  $M$ , this conclusion is weaker than simultaneous constant labels and is the correct statement. The distinction matters in the countable type argument of Section 13.

**Corollary 11.10** (Ultrapowers at the atomic stages [08 Cor. 8.7; 10 Prop. 12.1 at the full site]). *If the equivalent conditions of Theorem 7.5 hold, then*

$$\mathcal{S}_{M,\kappa}(F) \cong \prod_{\substack{U \in S_\kappa \\ U \supseteq F}} M^\omega / U.$$

*In particular, at the full stage,  $\mathcal{S}_{M,\mathfrak{c}^+}(F) \cong \prod_{U \in \omega^*, U \supseteq F} M^\omega / U$  and  $\text{Mix}_M(1) \cong \prod_{U \in \omega^*} M^\omega / U$ ; this recovers the product of all ultrapowers in the repository report.*

*Proof.* At the atom  $\{U\}$ , every mixture has only one nonzero Boolean piece. Its sequence labels are identified exactly by  $U$ -equality, so the value there is  $M^\omega / U$ . Atomicity says that the atoms below  $\sigma_F$  cover it. Distinct atoms have zero overlap, so arbitrary choices of these local sections glue uniquely. This is precisely the displayed product. Restriction to  $\sigma_F$  keeps exactly the ultrafilters extending  $F$ .  $\square$

Source 10 proves the full-site case by the same argument (a partition into singleton atoms assigns one raw label to each ultrafilter). Source 08 proves it at every atomic stage. This is the product formula already proved in the repository [1, Theorem 9.2]; the argument here identifies it with the uniform mixing construction. It is not the associated-sheaf formula on the countably generated site, whose complete algebra is atomless rather than atomic.

## 12 First-order transfer, specializations, and a sharp infinitary separation

### 12.1 Boolean truth on mixtures

Let  $M$  now be a nonempty, set-sized first-order structure in a set-sized finitary language. Constants are evaluated by constant sequences, and functions pointwise on sequences and piecewise on mixtures. If finitely many mixtures have different partitions, intersect those partitions to obtain a common refinement. A finite common refinement has join 1, by repeated distribution of finite meet over arbitrary joins.

For a first-order formula  $\varphi(\bar{v})$  and a tuple of raw sequences  $\bar{f}$ , write

$$V_\varphi(\bar{f}) = \{n < \omega : M \models \varphi(\bar{f}(n))\}.$$

The intended value on a raw tuple is  $e_\kappa([V_\varphi(\bar{f})])$ . For mixed tuples, Boolean logical connectives use the operations of  $\mathbb{C}_\kappa$ , and quantifiers use joins and meets over the set of global mixtures. Atomic equality is (19); other atomic relations use the analogous piecewise formula. The construction belongs to the standard framework of full Boolean-valued models and mixing; Ulrich's treatment gives a convenient modern account of that framework and its ordinary ultrafilter specializations [6, Sections 2–3]. We prove the specific facts needed here.

**Theorem 12.1** (Finitary transfer with an attained existential value [08 Thm 9.1; 10 Thm 10.1]). *Suppose  $\bar{x}$  is a finite tuple of global mixtures written on a common partition  $(b_i)$  as  $\bar{x} = \sum_i b_i \cdot \bar{f}_i$ . Then every finitary first-order formula satisfies*

$$\|\varphi(\bar{x})\|_\kappa = \bigvee_i b_i \wedge e_\kappa([V_\varphi(\bar{f}_i)]). \quad (21)$$

*The value is independent of all representations and satisfies the Boolean-valued first-order semantic clauses. For every formula  $\exists v \psi(v, \bar{x})$ , there is a global mixture  $y$  such that*

$$\|\exists v \psi(v, \bar{x})\|_\kappa = \|\psi(y, \bar{x})\|_\kappa = \bigvee_{z \in \text{Mix}_M(1)} \|\psi(z, \bar{x})\|_\kappa. \quad (22)$$

*For raw sequences,  $\|\varphi(\bar{f})\|_\kappa = e_\kappa([V_\varphi(\bar{f})])$ . For a tuple  $\bar{a}$  from  $M$ , the value on its constant copy is 1 or 0 according as  $M \models \varphi(\bar{a})$  or not; in particular, every sentence true in  $M$  has value 1, and every sentence false in  $M$  has value 0.*

*Proof.* For terms and atomic formulas, the statement is built into the piecewise definitions. Equality substitution is valid because, on any piece where the coordinate sequences agree, the ordinary atomic statements in  $M$  agree. This proves well-definedness at the atomic level. Refining partitions leaves the displayed join unchanged.

Suppose the formula identity has been proved for  $\varphi$ . Complementing a value piecewise on a partition gives

$$\neg \bigvee_i (b_i \wedge a_i) = \bigvee_i (b_i \wedge \neg a_i).$$

This follows by intersecting both sides with each  $b_i$ , which join to 1. Since  $e_\kappa$  preserves finite Boolean operations, ordinary negation in  $M$  proves the identity for  $\neg\varphi$ . The proof for conjunction is the same after common refinement; the other finitary connectives follow.

For the existential step, suppose the identity holds for  $\psi$ . For each  $i$  and each  $n$ , choose  $g_i(n) \in M$  satisfying  $\psi(g_i(n), \bar{f}_i(n))$  whenever any witness exists; use a fixed default element of  $M$  otherwise. This is a set-indexed choice, permitted in ZFC. Then, as literal subsets of  $\omega$ ,

$$V_\psi(g_i, \bar{f}_i) = V_{\exists v \psi}(\bar{f}_i).$$

The mixture  $y = \sum_i b_i \cdot g_i$  therefore has, by the induction hypothesis, exactly the right-hand side of (21) for  $\exists v \psi$ .

For any other candidate mixture  $z$ , refine its partition with the  $b_i$ . On each refined cell, ordinary truth in  $M$  gives  $\psi(z(n), \bar{f}_i(n)) \Rightarrow \exists v \psi(v, \bar{f}_i(n))$ . The induction hypothesis bounds its value by that same right-hand side. Taking the join over all candidates proves both the existential clause and (22). Universal quantification follows by negation. This completes the induction and establishes representation independence for all formulas.

The raw formula and the assertion about constants are immediate from their one-cell presentations. For a sentence, the set  $V_\varphi$  is either  $\omega$  or  $\emptyset$ , independently of the partition. Its value is therefore 1 or 0 as asserted.  $\square$

The theorem gives an exact internal, Boolean-valued first-order model of the theory of  $M$ . It does *not* say that the set of global sections, equipped with the relation “has Boolean value 1,” is an ordinary classical model of every sentence of that theory. For example, a Boolean disjunction can have value 1 without either disjunct having value 1. The field example in Section 14 makes this distinction concrete.

The maximum principle here also does not prove saturation for infinite types. It produces a witness for one existential first-order formula. Replacing a single formula by an infinite family introduces complete meets, which can change with the filter stage. Countable types are treated by a different argument, from source 10, in Section 13.

*Remark 12.2* (No illicit complete extension is used [10 Remark 10.2]). The proof never commutes  $e_\kappa$  with an arbitrary join in  $\mathbb{B}$ . Such a step would be unavailable, and above  $\mathfrak{p}$  it would conflict with Corollary 4.5. Existential truth is handled instead by one pointwise witness on each presentation cell. This is the precise reason fullness and transfer survive the failure of a complete comparison map.

## 12.2 Ordinary specializations

**Proposition 12.3** (Ordinary specializations [10 Prop. 10.3]). *Let  $\mathcal{V}$  be an ordinary ultrafilter on  $\mathbb{C}_\kappa$ . Quotient global mixtures by*

$$s \equiv_{\mathcal{V}} t \iff \|s = t\|_\kappa \in \mathcal{V}.$$

*With the induced functions and relations, the quotient  $M_{\kappa, \mathcal{V}}$  is an ordinary structure satisfying*

$$M_{\kappa, \mathcal{V}} \models \varphi([\bar{s}]_{\mathcal{V}}) \iff \|\varphi(\bar{s})\|_\kappa \in \mathcal{V}. \quad (23)$$

*The constant copy of  $M$  is an elementary substructure of this quotient.*

*Proof.* The Boolean equality laws and substitution make the quotient operations and relations well defined. Prove (23) by induction. Finite Boolean operations are decided by an ultrafilter. If an existential truth value belongs to  $\mathcal{V}$ , fullness in Theorem 12.1 supplies a witness with that same value, giving a witness in the quotient. The converse follows by monotonicity of truth under existential quantification. The claim about constants follows from their 0–1 truth values; distinct constants have equality value zero and hence remain distinct in the quotient.  $\square$

The ultrafilter  $\mathcal{V}$  in this proposition need not be a complete point of  $\mathbb{C}_\kappa$  (Remark 7.4). In particular, the proposition applies when  $\kappa \leq \mathfrak{u}$  and the Boolean locale has no complete points at all. It uses only Theorem 12.1 and therefore holds at every stage; source 10 states it at its stages.

### 12.3 An explicit infinitary formula detecting the threshold

**Theorem 12.4** (A  $\mathfrak{p}$ -fold conjunction separates the stages [08 Thm 9.2]). *For every  $\kappa > \mathfrak{p}$ , there are a structure  $M$ , a tuple consisting of one fixed old sequence  $f$ , and an infinitary formula  $\Psi(v) \in L_{\mathfrak{p}^+, \omega}$  such that*

$$\|\Psi(f)\|_\omega = 0, \quad \|\Psi(f)\|_\kappa > 0.$$

*Every finitary subformula has its old event as the Boolean value at both stages. Among conjunctions of fixed old events whose value changes from zero to positive, the arity  $\mathfrak{p}$  is minimal.*

*Proof.* Choose an SFIP family  $(A_\alpha)_{\alpha < \mathfrak{p}}$  with no infinite pseudointersection. Take the underlying set of  $M$  to be  $\omega$  and give the language unary predicates  $P_\alpha$ , interpreted as  $A_\alpha$ . Let  $f(n) = n$ , and form

$$\Psi(v) = \bigwedge_{\alpha < \mathfrak{p}} P_\alpha(v).$$

Interpret this infinitary conjunction as the complete Boolean meet of the values of its conjuncts. The individual values are  $e_\mu([A_\alpha])$  at stage  $\mu$ , by Theorem 12.1. Equation (11) therefore gives the two claimed values, the second being  $K_{\langle \text{Fr} \cup \{A_\alpha : \alpha < \mathfrak{p}\} \rangle}$ . The minimality assertion is Proposition 4.9.  $\square$

There is no contradiction with ordinary truth in  $M$ . The intersection of all the  $A_\alpha$  is finite, since an infinite intersection would itself be a pseudointersection. Thus the pointwise event for the whole infinitary conjunction is zero modulo finite sets. At the larger stage its *Boolean-valued* infinitary interpretation is positive. Theorem 12.1 was explicitly finitary; applying its formula to this infinitary conjunction would be invalid.

This distinction also explains the complete-homomorphism obstruction model-theoretically. A complete comparison preserving all old atomic predicates would preserve their conjunction. The two values prove that such a comparison cannot exist. We have not changed the interpretation of a finitary predicate, and we have not proved anything about the realization of every type of cardinality  $\mathfrak{p}$ .

### 12.4 The stable range includes the whole sheaf model

**Corollary 12.5** ([08 Cor. 9.3]). *When  $\kappa \leq \mathfrak{p}$ , the unique complete isomorphism  $\mathbb{C}_\omega \cong \mathbb{C}_\kappa$  fixing  $\mathbb{B}$  transports the mixing model at one stage to the mixing model at the other, preserving all old sequences and all Boolean truth values for which the semantics is defined by complete Boolean operations.*

*Proof.* Apply the isomorphism to every partition coefficient in a mixture, leaving its sequence labels unchanged. The equality formula, Boolean operations, and arbitrary joins and meets are preserved. The inverse isomorphism gives the inverse transport. In particular, the finitary truth formula and the explicitly defined infinitary conjunction semantics commute with transport.  $\square$

This gives more than an equality of abstract subterminal algebras: within the stable range the associated models themselves have the same explicit mixing description. At a small filter  $F$ , the support is represented in the original completion by  $\text{int}_\tau K_F$ , as in Proposition 4.10.

## 13 Localized realization and saturated specializations

This section is source 10's. Its results use Proposition 11.9, hence Theorem 5.2, and are stated at the successor stages  $\kappa = \lambda^+$ . Throughout,  $\lambda$  is an infinite cardinal and all mixtures, truth values and specializations are those of stage  $\lambda^+$ .

**Theorem 13.1** (Localized countable realization [10 Thm 11.1]). *Let  $w \in \mathbb{C}_{\lambda^+}$ , and let*

$$(\varphi_n(x, \bar{a}_n))_{n < \omega}$$

*be a countable family of formulas with finite tuples of mixed parameters  $\bar{a}_n$  from  $\text{Mix}_M(1)$ . Suppose that for every  $m < \omega$ ,*

$$w \leq \left\| \exists x \bigwedge_{n < m} \varphi_n(x, \bar{a}_n) \right\|. \quad (24)$$

*There is a mixed element  $s$  such that*

$$w \leq \|\varphi_n(s, \bar{a}_n)\| \quad (n < \omega). \quad (25)$$

*Proof.* The case  $w = 0$  is immediate. The countably many parameters can be simultaneously represented by raw sequences on a Boolean partition, by Proposition 11.9. Intersect this partition with  $w$  and discard zero cells. On a resulting cell  $c$ , write  $\tilde{f}_n$  for the raw tuple labeling  $\bar{a}_n$ , and define

$$\Sigma_m = \{k < \omega : M \models \exists x \bigwedge_{n < m} \varphi_n(x, \tilde{f}_n(k))\}.$$

These sets decrease with  $m$ , with  $\Sigma_0 = \omega$ . Equations (24) and (21), restricted to  $c$ , give

$$c \leq e_{\lambda^+}([\Sigma_m]) \quad (m < \omega). \quad (26)$$

At coordinate  $k$ , let

$$\ell(k) = \max(\{0\} \cup \{m \leq k : k \in \Sigma_m\}).$$

Choose  $g_c(k)$  satisfying the first  $\ell(k)$  formulas; such a witness exists by the definition of  $\Sigma_{\ell(k)}$ . If  $\ell(k) = 0$ , choose any fixed member of  $M$ . For a fixed  $n$ , whenever  $k \geq n + 1$  and  $k \in \Sigma_{n+1}$ , one has  $\ell(k) \geq n + 1$  and hence  $M \models \varphi_n(g_c(k), \tilde{f}_n(k))$ . Therefore

$$\Sigma_{n+1} \setminus \{0, \dots, n\} \subseteq V_{\varphi_n}(g_c, \tilde{f}_n). \quad (27)$$

Finite sets disappear in  $\mathbb{B}$ , so (26) and (27) imply  $c \leq e_{\lambda^+}([V_{\varphi_n}(g_c, \tilde{f}_n)])$  for every  $n$ .

Mix the sequences  $g_c$  over the partition of  $w$ , and use a default label outside  $w$ . On every cell the desired truth values contain that cell, so their global values contain  $w$ . This proves (25).  $\square$

For  $w = 1$  this is a uniform realization theorem for countable types whose finite parts are satisfiable with Boolean value 1. The local form is strictly more useful for specialization, because finite satisfiability in an ultrafilter quotient usually gives values *in the ultrafilter*, not values equal to 1.

**Theorem 13.2** (Every specialization is countably saturated [10 Thm 11.2]). *For every ordinary ultrafilter  $\mathcal{V}$  on  $\mathbb{C}_{\lambda^+}$ , the specialization  $M_{\lambda^+, \mathcal{V}}$  realizes every countable finitely satisfiable one-variable type with parameters from the specialization. In particular, when the language is countable,  $M_{\lambda^+, \mathcal{V}}$  is  $\aleph_1$ -saturated.*

*Proof.* Choose mixed representatives of the parameters in a countable type  $(\varphi_n(x, \bar{a}_n))_{n < \omega}$  and put

$$\theta_m = \left\| \exists x \bigwedge_{n < m} \varphi_n(x, \bar{a}_n) \right\|, \quad \theta_0 = 1.$$

Finite satisfiability in the quotient and Proposition 12.3 say that  $\theta_m \in \mathcal{V}$  for every  $m$ . These values form a decreasing sequence. Consider the Boolean partition, after omitting zero cells,

$$\pi_m = \theta_m \wedge \neg\theta_{m+1} \quad (m < \omega), \quad \pi_\infty = \bigwedge_{m < \omega} \theta_m. \quad (28)$$

Its join is 1: the finite differences exhaust the complement of the infinite meet. More generally, for each  $k$ ,

$$\theta_k = \pi_\infty \vee \bigvee_{m \geq k} \pi_m. \quad (29)$$

These identities hold in every complete Boolean algebra, by taking complements of the decreasing meets.

On  $\pi_m$ , fullness supplies an element realizing the first  $m$  formulas there. On  $\pi_\infty$ , Theorem 13.1 supplies one element realizing every formula there, since  $\pi_\infty \leq \theta_m$  for all  $m$ . Mix these elements along (28), using Lemma 11.3, to obtain  $s$ . For a fixed  $n$ , the formula  $\varphi_n(s, \bar{a}_n)$  holds on  $\pi_\infty$  and on every  $\pi_m$  with  $m \geq n + 1$ . Equation (29) therefore gives

$$\|\varphi_n(s, \bar{a}_n)\| \geq \theta_{n+1} \in \mathcal{V}.$$

By (23), the class  $[s]_{\mathcal{V}}$  realizes the whole type. In a countable language, all formulas in one variable over a countable parameter set form a countable set, so this proves the usual  $\aleph_1$ -saturation criterion.  $\square$

*Merge note.* By Corollaries 4.6 and 12.5, the stage- $\omega$  mixing model is the stage- $\aleph_1$  model transported along a marked isomorphism, so Theorems 13.1 and 13.2 hold for  $\mathbb{C}_\omega$  as well. They are not claimed at limit stages (Section 5.1). Theorem 13.2 answers the “useful first target” of source 08’s question Q6, “a precise theorem for countable types in a countable language, with the meaning of Boolean finite satisfiability fixed in advance”: finite satisfiability is read as membership of the values  $\theta_m$  in  $\mathcal{V}$  (or, in the localized form, as  $\theta_m \geq w$ ). Q6 is re-scoped in Section 19.

### 13.1 Scope of the model-theoretic conclusion

Theorem 13.2 is proved for the specific completed Fréchet models; it is not a general assertion that arbitrary Boolean-valued saturation survives arbitrary specialization. The localized diagonal construction and the failure-level partition (28) are the two ingredients that make the assertion work here. Nor is any higher saturation cardinal claimed. For an uncountable language the theorem still realizes countable types, but that assertion alone is not the full  $\aleph_1$ -saturation condition for all types over countable parameter sets.

The result also clarifies the role of  $\mathfrak{h}$ . A first failure of uncountable local joint decisions does not obstruct the countable refinements used in Theorem 13.1. All successor stages, and  $\mathbb{C}_\omega$ , remain countably distributive (Theorem 5.2), even when their complete algebras have different atoms and admit no event-preserving complete comparison.

## 14 Global rings: regularity and the idempotent spectrum

Let  $\mathbb{F}$  be a field with  $0 \neq 1$ , and let

$$A_{\mathbb{F}, \kappa} = \mathcal{S}_{\mathbb{F}, \kappa}(\text{Fr})$$

be the ring of global sections. Ring operations are the piecewise operations from the mixing construction.

**Theorem 14.1** (Regularity and Boolean idempotents [08 Thm 10.1]). *The ring  $A_{\mathbb{F}, \kappa}$  is a commutative von Neumann regular ring: for every  $x$  there is  $y$  with  $xyx = x$ . Its Boolean algebra of idempotents is canonically  $\mathbb{C}_\kappa$ :*

$$\text{Idem}(A_{\mathbb{F}, \kappa}) \cong \mathbb{C}_\kappa. \quad (30)$$

Under this identification, primitive nonzero idempotents correspond exactly to the ultrafilters  $U$  with  $\chi_*(U) < \kappa$ .

*Proof.* Write  $x = \sum_i b_i \cdot f_i$ . Define the generalized inverse sequence

$$g_i(n) = \begin{cases} 0, & f_i(n) = 0, \\ f_i(n)^{-1}, & f_i(n) \neq 0. \end{cases}$$

The mixture  $y = \sum_i b_i \cdot g_i$  satisfies  $xyx = x$  pointwise on every piece, hence globally.

For  $b \in \mathbb{C}_\kappa$ , the element  $\epsilon_b = b \cdot 1 + (-b) \cdot 0$  is idempotent. The map  $b \mapsto \epsilon_b$  is injective because its value-1 event is exactly  $b$ . Products, Boolean complements, and Boolean joins of these idempotents are respectively

$$\epsilon_b \epsilon_c = \epsilon_{b \wedge c}, \quad 1 - \epsilon_b = \epsilon_{-b}, \quad \epsilon_b + \epsilon_c - \epsilon_b \epsilon_c = \epsilon_{b \vee c}.$$

Conversely suppose  $x^2 = x$ . In a field,  $t^2 = t$  is equivalent to  $t = 0 \vee t = 1$ . By Theorem 12.1,  $\|x = 0\| \vee \|x = 1\| = 1$ . The two events are disjoint, since  $0 \neq 1$ . Therefore  $x$  agrees with 0 on one piece and with 1 on the other; the equality formula shows  $x = \epsilon_{\|x=1\|}$ . This proves surjectivity. An idempotent is primitive exactly when it is nonzero and has no nonzero strictly smaller idempotent in the Boolean order. Hence primitive idempotents are atoms, now classified by Theorem 7.1.  $\square$

**Corollary 14.2** (A non-product theorem for the countable site [08 Cor. 10.2]). *For every field  $\mathbb{F}$ , if  $\kappa \leq \mathfrak{u}$ , the global ring  $A_{\mathbb{F},\kappa}$  is not isomorphic to a full direct product of nonzero fields. In particular, the associated global ring on the countably generated free-filter site is not a product of ultrapower fields. At an atomic stage, by contrast,*

$$A_{\mathbb{F},\kappa} \cong \prod_{U \in S_\kappa} \mathbb{F}^\omega / U.$$

*Proof.* A nonempty product of nonzero fields has primitive idempotents, one for each coordinate. The ring here has none when  $\kappa \leq \mathfrak{u}$ , by Theorem 14.1. The empty product is the zero ring and is excluded because  $A_{\mathbb{F},\kappa}$  has distinct constant sections 0, 1. The atomic-stage formula is Corollary 11.10.  $\square$

This is a statement about a *full direct product*; it does not deny familiar subdirect-product representations of reduced rings. Nor does it assert that the global ring is a field in the atomless case. In fact  $\mathbb{B}$  has nontrivial events, so (30) supplies nontrivial idempotents at every stage. For  $\mathbb{F} = \mathbb{R}$ , the sheaf is internally an ordered field model with Boolean truth, while its global ring is not a single hyperreal field. Source 10 makes the same observation (its Example 11.3): the completed object satisfies the field axioms with Boolean value 1, every specialization of Proposition 12.3 is a field, and yet, for a nontrivial region  $b$ , the mixture of 1 on  $b$  and 0 on  $\neg b$  is an idempotent distinct from both global constants. This is not a counterexample to transfer: being one of the two constants holds with Boolean value 1 as a disjunction, while neither equality separately has value 1.

The ring also remembers the marked-completion obstruction. Any ring map preserves finite Boolean operations on idempotents. If a ring comparison between the two stages were required additionally to preserve all joins of idempotents and to fix the characteristic sequences of subsets of  $\omega$ , its restriction to idempotents would be the forbidden complete Boolean map of Corollary 4.5, in either direction. Thus above  $\mathfrak{p}$  no comparison with these combined requirements exists. Ordinary ring homomorphisms need not preserve arbitrary idempotent joins; without that requirement this argument gives no prohibition.

## 15 The two endpoints and their exact cardinalities

This section is source 10's. By the index shift, its  $S_{\aleph_0}(A)$  is  $\text{Mix}_M(1)$  at stage  $\aleph_1$ , and its  $S_{\mathfrak{c}}(A)$  is  $\text{Mix}_M(1)$  at the full stage  $\mathfrak{c}^+$ .

**Lemma 15.1** ([10 Lemma 12.2]). *One has  $|\mathbb{B}| = \mathfrak{c}$  and  $|\mathbb{C}_\omega| = 2^\mathfrak{c}$ . Every antichain in  $\mathbb{C}_\omega$  has size at most  $\mathfrak{c}$ .*

*Proof.* Every equivalence class of subsets of  $\omega$  modulo finite difference is countable. Since there are  $\mathfrak{c}$  subsets, their quotient has cardinality  $\mathfrak{c}$ . A regular open of  $\mathbb{B}^+$  is a subset of a set of size  $\mathfrak{c}$ , so  $|\mathbb{C}_\omega| \leq 2^\mathfrak{c}$ .

For the opposite inequality, identify the countable set  $2^{<\omega}$  of finite binary strings with  $\omega$ . For each real  $x \in 2^\omega$ , take the infinite set of its initial segments. Distinct branches have only finitely many common initial segments. Their classes therefore form an antichain of size  $\mathfrak{c}$  in  $\mathbb{B}^+$ . In the completion, joins of distinct subfamilies of this antichain are distinct: an antichain member in their symmetric difference is below one join and disjoint from the other. Thus  $|\mathbb{C}_\omega| \geq 2^\mathfrak{c}$ .

Finally, below each member of an antichain in  $\mathbb{C}_\omega$  choose a nonzero element of the order-dense copy of  $\mathbb{B}$ . The chosen elements must be distinct. This injects the antichain into a set of size  $\mathfrak{c}$ .  $\square$

**Theorem 15.2** (Three exact sizes [10 Thm 12.4]). *For every set  $M$  with  $2 \leq |M| \leq \mathfrak{c}$ ,*

$$|\mathcal{R}_M(\text{Fr})| = \mathfrak{c}, \quad (31)$$

$$|\mathcal{S}_{M, \aleph_1}(\text{Fr})| = 2^\mathfrak{c}, \quad (32)$$

$$|\mathcal{S}_{M, \mathfrak{c}^+}(\text{Fr})| = 2^{2^\mathfrak{c}}. \quad (33)$$

*Proof.* There are at most  $|M|^{\aleph_0} \leq \mathfrak{c}$  raw sequences. Fixing two distinct labels embeds binary sequences modulo finite difference into  $\mathcal{R}_M(\text{Fr})$ , giving the lower bound  $|\mathbb{B}| = \mathfrak{c}$  in (31).

At the countably generated stage the complete algebra is  $\mathbb{C}_\omega$  (Corollary 4.6). A mixed presentation uses an antichain of size at most  $\mathfrak{c}$ , by Lemma 15.1, with at most  $\mathfrak{c}$  possible raw labels. There are at most

$$(2^\mathfrak{c})^\mathfrak{c} \cdot \mathfrak{c}^\mathfrak{c} = 2^\mathfrak{c}$$

such presentations, so there are at most that many quotient elements. Conversely, mix two fixed distinct constant labels along  $b, -b$  for every  $b \in \mathbb{C}_\omega$ . Distinct  $b$  give distinct sections, so the global sections number at least  $|\mathbb{C}_\omega| = 2^\mathfrak{c}$ .

At the full-site stage, Corollary 11.10 gives a product of  $|\omega^*| = 2^\mathfrak{c}$  factors (Lemma 9.2), each of size at least 2 and at most  $\mathfrak{c}$ . Consequently its size lies between  $2^{2^\mathfrak{c}}$  and

$$\mathfrak{c}^{2^\mathfrak{c}} = (2^{\aleph_0})^{2^\mathfrak{c}} = 2^{2^\mathfrak{c}}.$$

This proves (33).  $\square$

*Merge note.* By Corollary 12.5, the global sections at every stage  $\kappa \leq \mathfrak{p}$ , in particular at stage  $\omega$ , are in bijection with those at stage  $\aleph_1$ , so (32) gives  $2^\mathfrak{c}$  there too.

The increases are strict by Cantor's theorem. They do not depend on CH. The first increase is the cost of allowing arbitrary Boolean mixing along the atomless completion. The second is the cost of allowing independent choices at every free ultrafilter. These are cardinal comparisons, not a construction of complete embeddings between the two completed truth-value algebras. Corollary 4.5 excludes the canonical complete embeddings one might otherwise expect.

## 16 General Boolean algebras and intermediate comparisons

The representation theorem is not peculiar to  $\mathcal{P}(\omega)/\text{Fin}$ . Let  $B$  be any nontrivial Boolean algebra. For each infinite  $\kappa$ , take all proper filters of  $B$  generated by fewer than  $\kappa$  elements, and put on  $\text{Ult}(B)$  the topology generated by their sets of ultrafilter extensions. The proof of Theorem 3.3 applies verbatim. This supplies a convenient general mechanism whose specialized thresholds can be computed from the algebra itself.

**Definition 16.1** ([08 Def. 11.1]). Let  $\mathfrak{p}(B)$  be the least size of a centered family in  $B^+$  with no nonzero common lower bound, and set  $\mathfrak{p}(B) = \infty$  if no such family exists. Here centered means that every finite meet is nonzero.

**Proposition 16.2** (The abstract threshold mechanism [08 Prop. 11.2]). *For the filter hierarchy of  $B$ , the original  $B^+$  is order dense at stage  $\kappa$  exactly when every centered family of size below  $\kappa$  has a nonzero lower bound. Equivalently, this holds when  $\kappa \leq \mathfrak{p}(B)$ , interpreting  $\infty$  as larger than every cardinal. When it fails, a centered family witnessing the failure gives a zero-to-positive meet and obstructs every complete homomorphism from the original completion that fixes  $B$ .*

*Proof.* A principal filter generated by  $b > 0$  extends the filter generated by  $\mathcal{A}$  exactly when  $b$  is a lower bound for  $\mathcal{A}$ . Thus the first assertion is the order-density criterion. For a centered family with no positive lower bound, the intersection of its Stone clopens is nonempty by the ultrafilter lemma but has empty original interior. Once the family is small enough to generate an allowed filter, that intersection is open at the new stage. The two meet calculations are exactly those of (11).  $\square$

For  $B = \mathcal{P}(\omega)/\text{Fin}$  this invariant is the usual  $\mathfrak{p}$ . For the finite-cofinite algebra of Section 10, it is  $\aleph_0$ . For a finite Boolean algebra it is  $\infty$ . The construction therefore explains both examples within one proof, without claiming that this abstract reformulation defines an unprecedented invariant. Proposition 16.2 concerns only the forward direction; an analogue of the reverse barrier (Theorem 4.4) for  $B$  would also need a density theorem for filters with no nonzero lower bound, as source 10 points out in its question on other ideals (R9 in Section 19).

**Proposition 16.3** (A criterion for comparing arbitrary stages [08 Prop. 11.3]). *Let  $\kappa \leq \mu$  be infinite cardinals. The inclusion of filter posets  $\mathbb{P}_\kappa \subseteq \mathbb{P}_\mu$  is order dense exactly when*

$$\forall F \in \mathbb{P}_\mu \exists G \in \mathbb{P}_\kappa \quad G \supseteq F. \quad (34)$$

*When this holds, their completions are canonically completely isomorphic in a way respecting every condition from  $\mathbb{P}_\kappa$ , and their reduced-power mixing models agree under this isomorphism.*

*Proof.* The displayed condition is precisely order density. Compatibility of conditions of  $\mathbb{P}_\kappa$  is the same in both posets by Lemma 2.4; hence a dense inclusion gives the unique common completion. A raw sequence-equality event is represented by a condition from the finite-generation stage, so these events are fixed. Transporting partition coefficients then identifies the mixing constructions as in Corollary 12.5.  $\square$

The converse in this proposition is asserted for *order density of the inclusion*, not for the existence of an arbitrary abstract isomorphism. A single complete algebra can have many inequivalent presentations and many different embeddings of a common subalgebra.

## 17 A consolidated picture and worked regimes

The following table, from source 10, is translated into the indexing of this report.

| Boundary       | Exact property                                                                                                                                         | Limitation or further distinction                                                                                                               |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| $\mathfrak{p}$ | For $\kappa \leq \mathfrak{p}$ the old events are order dense; for $\kappa > \mathfrak{p}$ neither complete comparison preserving those events exists. | This alone is not an unlabeled nonisomorphism theorem.                                                                                          |
| $\mathfrak{h}$ | First failure of local joint decisions for binary sections of the stage- $\aleph_1$ (equivalently stage- $\omega$ ) sheaf.                             | At successor stages $\kappa > \mathfrak{h}$ , partition distributivity also separates $\mathbb{C}_\kappa$ abstractly from $\mathbb{C}_\omega$ . |
| $\mathfrak{u}$ | First existence of atoms or complete Boolean points in $\mathbb{C}_\kappa$ , at $\kappa = \mathfrak{u}^+$ .                                            | Some atoms do not imply that the entire algebra is atomic.                                                                                      |
| $\mathfrak{c}$ | At $\kappa = \mathfrak{c}^+$ every free filter is admitted, so the algebra is $\mathcal{P}(\omega^*)$ and the sheaf is a product of ultrapowers.       | The least bound giving a wholly atomic algebra need not be determined by the preceding point-existence statement.                               |

### 17.1 Under the continuum hypothesis

If  $\mathfrak{c} = \aleph_1$ , the inequalities proved above give  $\mathfrak{p} = \mathfrak{h} = \mathfrak{u} = \mathfrak{c} = \aleph_1$ . There are then only two different completions at the infinite cardinal stages: the stages  $\omega$  and  $\aleph_1$  share the countably generated completion  $\mathbb{C}_\omega$  (Corollary 4.6), and every stage  $\kappa \geq \aleph_2$  is the full stage  $\mathbb{P}_{\mathfrak{c}^+}$ . The former has atomless algebra  $\mathbb{C}_\omega$  and  $2^{\aleph_1}$  completed binary sections; the latter has the power set of  $2^{\aleph_1}$  ultrafilters and  $2^{2^{\aleph_1}}$  completed binary sections. There is no complete event-preserving comparison in either direction. This is a conditional example, not an assumption used elsewhere. In source 10's indexing these are its two stages  $\mathcal{F}_{\aleph_0}$  and  $\mathcal{F}_{\aleph_1} = \mathcal{F}_{\mathfrak{c}}$ .

### 17.2 When the characteristics separate

If  $\mathfrak{p} = \mathfrak{c}$ , then all stages  $\kappa \leq \mathfrak{c}$  yield the same completion over  $\mathbb{B}$ , by Theorem 4.3. The inequalities also force  $\mathfrak{h} = \mathfrak{u} = \mathfrak{c}$ .

If  $\mathfrak{p} < \mathfrak{u}$ , every stage in the interval  $\mathfrak{p} < \kappa \leq \mathfrak{u}$  produces an atomless algebra for which the two-sided canonical comparison already fails. Thus appearance of atoms cannot be the mechanism responsible for the first failure. If  $\mathfrak{p} < \mathfrak{h}$ , the labeled obstruction appears before the particular unlabeled distributivity obstruction. These are implications under the displayed inequalities; no forcing consistency result about separating the characteristics is needed for them, and none is claimed.

### 17.3 Why ordinary model theory does not see everything

For a fixed  $M$ , all the specializations  $M_{\kappa, \mathcal{V}}$  satisfy the same first-order theory as  $M$ , and at the successor stages, in a countable language, they are countably saturated. Yet the Boolean algebras before specialization have different point behavior, different dense joins, and potentially very different global section cardinalities. Elementary equivalence and countable saturation therefore do not classify the pre-specialized objects.

This is not a paradox. Specialization decides Boolean events using an ordinary ultrafilter and forgets most of the completion structure. Corollary 4.5 concerns preservation of *all* joins while keeping the original events fixed, a demand far stronger than ordinary first-order truth preservation.

## 18 Proof audit and reproducible finite checks

### 18.1 Foundational and logical boundaries

The results separate naturally into three groups. The Stone/filter representation, the comparison theorems in both directions, the closure,  $\mathfrak{h}$ , atom and point results, the forcing reading, the mixing construction, transfer, specialization and saturation, the ring consequences and the exact sizes are proved here in ZFC. The properties of the forcing model in Proposition 9.1 are imported from Millán; the three-stage consequence is proved from those properties. The source report supplies the motivating problem and its previous gluing result, but no theorem in this report depends on trusting its unformalized proof. The set-theoretic results do not depend on a large-cardinal hypothesis, on CH, on a reported Lean theorem, or on a conjectural value of a cardinal characteristic. The proofs of the main comparison statements are self-contained relative to elementary complete Boolean algebra, and the model construction is given explicitly so that the uses of mixing, choice, and countable refinement can be inspected separately.

Choice is used at visible points: ultrafilter extension; selection of bases and representatives; maximal antichain refinements; pointwise witnesses in a general structure; independent ultrafilter choices; and cardinal arithmetic arguments. This report does not extend the source report’s choice-free guarantees to the new theory. Some parts require less than full choice, but an exact reverse-mathematical classification is left as a further task (Q8, R8).

The most important negative checks of source 08 are the following.

- A1. **The bound is strict in the definition of the site.** The original completion persists at  $\kappa = \mathfrak{p}$ ; change occurs at  $\mathfrak{p}^+$ . Atoms first occur at  $\mathfrak{u}^+$ , not at  $\mathfrak{u}$ . (Source 10’s “at and above  $\mathfrak{p}$ ” and “atoms at  $\mathfrak{u}$ ” are the same statements in its non-strict indexing; see Section 1.6.)
- A2. **A marked obstruction is not an abstract classification.** The no-homomorphism theorems fix the embedded  $\mathbb{B}$ . Proposition 10.1 shows why the qualifier is indispensable.
- A3. **An atom is not atomicity.** The latter requires (34) with ultrafilter targets, as stated in Theorem 7.5.
- A4. **A Boolean point is not an ordinary ultrafilter.** A locale point preserves arbitrary joins. This is why the countable-site Booleanization is pointless.
- A5. **Finitary transfer is not infinitary transfer or saturation.** The explicit  $\mathfrak{p}$ -conjunction distinguishes these notions.
- A6. **An internal field need not have a field of global sections.** The idempotent algebra is a direct obstruction to that inference.
- A7. **Regularity of  $\kappa$  is used only where stated.** The topology and threshold theorems allow singular cardinals; the closure proof assumes regularity, and source 10’s closure consequences are stated only at the successor stages.

Source 10’s checks against tempting invalid arguments, which its proofs use rather than add afterwards, are these.

- B1. Boolean joins of membership events are regularized unions.
- B2. All complete comparison maps are required to be unital.
- B3. The reverse obstruction uses density of the bad filters, not just the existence of one.
- B4. Complete locale points are not ordinary Boolean ultrafilters.

- B5. Infinite-valued mixed elements require sequence labels.
- B6. Specialization saturation is proved using (28), not inferred from Boolean-1 finite satisfiability alone.
- B7. A character bound is on an ordinary filter base; all cardinal multiplication arguments use an infinite cardinal.

The merge adds two: the index shift of Lemma 2.2 is applied to every imported statement, and no closure-based statement of source 10 is printed for a stage where its closure input is unproved (Section 5.1).

## 18.2 Dependencies of the principal results

| Result                                 | Essential ingredients used here                                                                                               |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| Representation, Theorem 3.3            | Ultrafilter extension; the clopen base $K_F$ ; regularization.                                                                |
| General completion criterion           | Positive adjunction; regular cones; order-dense completion; the dense incompatible family $\mathcal{E}_F$ .                   |
| Marked $\mathfrak{p}$ threshold        | The definition of $\mathfrak{p}$ ; pseudointersections of small generating families; the two meets of one SFIP family.        |
| Reverse barrier                        | The criterion; transporting an SFIP family into a pseudointersection; density of no-pseudointersection filters.               |
| $\mathfrak{h}$ local-reading boundary  | Dense decision sets; maximal antichains; distinct real codes for at most $\mathfrak{c}$ cells.                                |
| $\mathfrak{u}$ atom and point boundary | Splitting by a Stone clopen or by one set; complete Boolean decomposition; the kernel-join proof for points.                  |
| Forcing reading                        | Closure at successor stages; deciding names; density of conditions avoiding a fixed ultrafilter.                              |
| Explicit associated sheaf              | Boolean partitions; raw equality test; flattening; local representability; the universal gluing argument.                     |
| Transfer and specializations           | Finite common refinement; pointwise witnesses using choice; Boolean equality laws.                                            |
| Saturated specializations              | Countable partition refinement; pointwise witnesses; finite-cutoff diagonal; the Boolean failure-level partition.             |
| Early atomicity                        | The imported model properties; the atomicity criterion; counting generating families.                                         |
| Exact endpoint sizes                   | A continuum-sized almost disjoint family; an explicit independent family; counting mixed presentations; the product endpoint. |

These are preparation-stage proof checks, not independent refereeing or machine verification. Compilation and PDF inspection verify the artifact, not the theorems. No finite experiment is offered as certification of an infinite cardinal-threshold claim.

## 18.3 What the executable supplement checks

Source 08's package contains `verify_finite.py`, shipped with this report as `code/08-filter-complexity-verify_finite.py`, using only exact integer arithmetic and the Python standard library. It checks finite Boolean analogues, not the infinite cardinal theorems. For  $B = \mathcal{P}(n)$ , a proper filter is determined by its nonempty kernel  $K \subseteq n$ . The program verifies the filter/Stone order reversal, compatibility, relative dense sieves, and the resulting atom-supported coverage. It then checks the equality formula and existential-witness calculation on finite products, and checks the generalized inverse and all idempotents over  $\mathbb{F}_3$ .

These are tests of the algebraic interfaces used in the proofs. There is no finite experiment capable of verifying that a family witnesses  $\mathfrak{p}$ , that an ultrafilter has a particular uncountable character, or that the cited

forcing model exists. The recorded JSON output, shipped as `data/08-filter-complexity-verification_results.json`, is actual output of the supplied script, not a substitute for those proofs. The report README gives the rerun command, and source 08’s audit file `08-filter-complexity-proof_audit.md` records the scope of every check. Source 10 supplies no computational check. No Lean or Rocq verification is claimed.

The deterministic run supplied with source 08 produced:

| Check                                       | Number checked |
|---------------------------------------------|----------------|
| Filter order and compatibility pairs        | 284            |
| Relative filter families                    | 33,508         |
| Sieves among those families                 | 332            |
| Dense sieves among those sieves             | 189            |
| Candidate Boolean frame-point maps          | 65,812         |
| Mixture equality comparisons                | 507,348        |
| Pointwise generalized inverses              | 363            |
| Idempotent Boolean-operation pairs          | 1,364          |
| Existential instances / candidate witnesses | 360 / 22,140   |

The script also exhibits a nonzero nonunit in  $\mathbb{F}_3^2$  for which the internal field disjunction still has full Boolean value. This small example checks the internal/external distinction, not any statement about ultrafilters on an infinite set.

## 18.4 A formalization path

A practical formalization should begin with an arbitrary Boolean algebra and postpone  $\omega^*$  and cardinal arithmetic. First prove that proper filters correspond to nonempty Stone closed sets and that finite unions of small bases preserve compatibility. Next formalize the dense-site/regular-open representation and the principal-filter density criterion. The pseudointersection theorem then becomes an instantiation of a purely order-theoretic lemma.

The mixing construction can be formalized independently over an abstract complete Boolean algebra with a Boolean-valued sequence structure. The key finite-support proof obligations are equality transitivity, finite common refinement, and the existential witness lemma. Only after these are established should one connect the locale sheaf construction and the cardinal-indexed sites.

The relative-consistency application should remain a clearly named external hypothesis until its forcing input has actually been formalized. A file with an admitted statement is not the same as a checked derivation of that statement; the theorem dependency ledger must retain that boundary. Source 10 proposes a different first target, the two-sided barrier (question R10); both orders are recorded.

## 19 Further research questions and proposed projects

The questions below are a proposed research program, the union of the two sources’ lists. Except where a specific published problem has been cited, they are not claims that the literature has established them as open, and no claimed solution is hidden in their wording. The point is to state exact next targets and the results already available for attacking them. Questions Q1–Q10 are source 08’s; R1–R10 are source 10’s research questions 14.1–14.10, translated into this report’s indexing. Where one source answers or partly answers a question of the other, the question is re-scoped and the answer is named.

## 19.1 Source 08's questions

- Q1. Classify intermediate atomicity.** Determine the consistency possibilities for  $\text{GE}(S_\kappa, \kappa)$  when  $\mathfrak{u} < \kappa \leq \mathfrak{c}$ . Theorem 7.5 turns this into an exact extension question; Theorem 9.3 gives one positive intermediate example. Is its truth determined by familiar cardinal characteristics in any substantial range, or does it require more detailed ultrafilter-character information? (Parallel to R2.)
- Q2. Determine mixed atomic and atomless regions.** For specified  $\kappa$ , analyze when  $0 < a_\kappa < 1$  in (16). Equivalently, small ultrafilters exist but fail to occur above some allowed condition. Describe the relative completion on the complement of their regular-open support. The first step is a literature audit of small-character ultrafilter extension failures, not the unsupported assumption that  $\mathfrak{u} < \kappa$  already decides the issue. (Parallel to R3; Proposition 8.1 describes the residual part as the region where forcing adds a new ultrafilter.)
- Q3. Compute the full stabilization relation.** For arbitrary  $\kappa \leq \mu$ , decide the extension property (34). The special case  $\kappa = \omega$  is settled here exactly by  $\mathfrak{p}$ . Other pairs ask whether complicated filters can always be strengthened to filters with smaller bases. Determine whether atomicity, once reached, can be lost at a later stage; no monotonicity assertion follows from the theorems proved here. (Parallel to R4.)
- Q4. Separate marked and unmarked isomorphism problems.** Above  $\mathfrak{p}$  the completions cannot be compared by a complete map fixing all original events, in either direction. Determine when they can nevertheless be abstractly isomorphic, and which invariants distinguish them. Atom counts settle the two atomic stages in our model, while the finite-cofinite example shows that marking can matter even when atom counts coincide. Cellularities, distributivity spectra, and relative densities are natural additional tests. *Re-scoped:* Corollary 6.3 (source 10) settles the unmarked question negatively at every successor stage  $\kappa > \mathfrak{h}$ , using partition distributivity. What remains is the interval  $\mathfrak{p} < \kappa \leq \mathfrak{h}$  (R1) and the limit stages.
- Q5. Analyze singular stages.** At singular  $\kappa$ , compute closure and distributivity of  $\mathbb{P}_\kappa$  and its completion. The union-of-bases argument fails at a clearly identified point. One should distinguish the poset having a particular closure property from it possessing a dense closed presentation, and from distributivity of the complete Boolean algebra. Source 10's closure consequences (local decisions, the forcing reading, simultaneous raw representatives, countable saturation) are not proved at these stages (Section 5.1); deciding whether they hold there is part of this question.
- Q6. Develop a genuine type-realization theory.** Characterize the cardinals  $\mu$  for which relevant  $\mu$ -sized Boolean-valued types over the mixing model are realized. The maximum principle proved here handles one existential formula, and the  $\mathfrak{p}$ -conjunction gives a sharp fixed-parameter obstruction to preserving infinitary truth. *Re-scoped:* the question's "useful first target", a precise theorem for countable types in a countable language with the meaning of Boolean finite satisfiability fixed in advance, is settled by source 10's Theorems 13.1 and 13.2 at the successor stages and at stage  $\omega$ . What remains is types of size at least  $\aleph_1$  (higher saturation, R7), countable types in the Boolean-valued model itself beyond the localized form, and the limit stages.
- Q7. Replace the finite ideal.** Apply Proposition 16.2 to  $\mathcal{P}(I)/I$  for other ideals and index sets. Compute both the centered-family lower-bound threshold and the character spectrum of quotient ultrafilters. Distinguish a genuinely new calculation for a particular ideal from merely renaming a known ideal pseudointersection invariant. (Parallel to R9.)
- Q8. Remove unnecessary choice.** Develop a point-free version directly from regular downsets of the

filter poset, avoiding an a priori set of Stone points. Determine which comparison theorems require the Boolean prime ideal theorem, which use dependent choice, and which require stronger choice for representative selection. This would connect the present ZFC theory more tightly to the source report's choice-free gluing obstruction. (Parallel to R8.)

- Q9. Recover filter complexity from algebraic models.** The idempotent algebra of  $A_{\mathbb{F},\kappa}$  is exactly  $\mathbb{C}_\kappa$ . Determine what further information the ring retains about the marked original sequence algebra, its elementary ultrapower factors, and the stage parameter. Classify ring maps that preserve complete idempotent joins separately from ordinary ring homomorphisms, for which our obstruction is insufficient.
- Q10. Build a checked library with explicit assumptions.** Implement the dependency order in Section 18, beginning with the abstract Boolean statements. A realistic first milestone is a kernel-checked proof of the principal-filter density equivalence and the complete-homomorphism obstruction; a second is the general mixing sheaf. The forcing model should be a separately audited assumption until its proof is available. (Parallel to R10.)

## 19.2 Source 10's questions

- R1. Unlabeled isomorphism below  $\mathfrak{h}$**  (source 10, 14.1). Let  $\kappa = \lambda^+$  with  $\mathfrak{p} \leq \lambda < \mathfrak{h}$ , that is, a successor stage with  $\mathfrak{p} < \kappa \leq \mathfrak{h}$ . Can  $\mathbb{C}_\kappa$  be abstractly isomorphic to  $\mathbb{C}_\omega$ , despite the absence of any complete comparison preserving the original copy of  $\mathbb{B}$ ? Which additional invariants, such as density, antichain spectra, or homogeneity, decide this? Corollaries 4.5 and 6.3 leave exactly this distinction unresolved by the arguments given here. An answer should specify whether it concerns an abstract algebra, an algebra with its distinguished subalgebra, or the associated site; these are different classification problems. (Parallel to Q4.)
- R2. When are small-character ultrafilters dense?** (source 10, 14.2). For which infinite cardinals  $\lambda < \mathfrak{c}$  does every  $F \in \mathbb{P}_{\lambda^+}$  extend to some  $U \in S_{\lambda^+}$ ? Can this extension property hold strictly below  $\mathfrak{c}$ , and how does it depend on the ultrafilter character spectrum? By Theorem 7.5, this is exactly the question when  $\mathbb{C}_{\lambda^+}$  is wholly atomic. The existence of one small-character ultrafilter is insufficient for the proof. We do not assume that this extension property is monotone in  $\lambda$ : increasing the bound adds both possible ultrafilters and new filters requiring extensions. *Re-scoped*: the second clause is answered positively, relative to the imported model of Proposition 9.1, by source 08's Theorem 9.3: there  $\mathbb{P}_{\aleph_2}$  ( $\lambda = \aleph_1$ ) is wholly atomic while  $\aleph_1 < \mathfrak{c} = \aleph_2$ . This is a relative-consistency answer, not a ZFC theorem. The first and third clauses remain open. (Parallel to Q1.)
- R3. Residual invariants** (source 10, 14.3). When  $r_\kappa \neq 0$ , determine the density, cellularity, distributivity spectrum, and forcing equivalence class of  $\mathbb{C}_\kappa \upharpoonright r_\kappa$ . How much information about its isomorphism type is carried by the spectrum  $\{\chi(U) : U \in \omega^*\}$ ? Proposition 8.1 supplies a concrete constraint at successor stages  $\kappa = \lambda^+$ : the residual forcing adds a new ultrafilter of character greater than  $\lambda$  while adding no sequences of ground-model objects of length at most  $\lambda$ . It does not identify that forcing uniquely. (Parallel to Q2.)
- R4. Comparisons between two bounded stages** (source 10, 14.4). For  $\mathfrak{p} \leq \lambda < \mu \leq \mathfrak{c}$ , characterize complete Boolean homomorphisms between  $\mathbb{C}_{\lambda^+}$  and  $\mathbb{C}_{\mu^+}$  that preserve every event  $[A]$ . Is there a useful extension or approximation criterion on filters that replaces the pseudointersection criterion? The proof at the lowest stage works because an infinite set is exactly a nonzero element of an order-dense base. At higher stages that base has already lost order density, so applying the same argument without modification would beg the question. Source 08's Proposition 16.3 supplies a

sufficient criterion, order density of the inclusion, which partly addresses this question. (Parallel to Q3.)

- R5. The next distributivity failure** (source 10, 14.5). For a nonzero atomless residual, what is the least size of a family of Boolean partitions with no common dense refinement? At stage  $\lambda^+$ , Theorem 5.2 shows that this least size is at least  $\lambda^+$ . Can that bound be sharp, or can there be substantially more distributivity? The answer should distinguish partition distributivity from binary decision distributivity if the algebra has antichains too large to be coded by countably many bits. The real-code reduction in Theorem 6.2 uses the special antichain bound for  $\mathbb{C}_\omega$ .
- R6. Joint raw labels versus joint constant labels** (source 10, 14.6). For infinite  $M$ , determine the least size of a family of completed sections that cannot be simultaneously represented by raw sequences on a dense cover. How does this differ from the joint-constant threshold for finite-valued sections? Proposition 11.9 gives a lower bound at every successor stage, and Theorem 6.2 completely answers the binary constant-label version at the initial stage. The identity sequence in Example 11.8 prevents treating these as the same problem for infinite  $M$ .
- R7. Higher saturation** (source 10, 14.7). Which larger types are realized in  $M_{\lambda^+, \mathcal{V}}$ ? How do the theory of  $M$ , the ordinary ultrafilter  $\mathcal{V}$  on  $\mathbb{C}_{\lambda^+}$ , and the partition distributivity of  $\mathbb{C}_{\lambda^+}$  interact in determining saturation above  $\aleph_1$ ? Theorem 13.2 settles the countable case uniformly. Its coordinate diagonal uses the finite cutoff  $\ell(k) \leq k$  and cannot simply be stretched to uncountably many requirements. Higher saturation should be compared carefully with the notions for full Boolean-valued models and specializations in [6]. (Parallel to the remainder of Q6.)
- R8. Choice strength** (source 10, 14.8). Which parts of the hierarchy and its comparison barriers can be proved in ZF, ZF + DC, or with only the Boolean prime ideal theorem? What is the exact choice strength of the explicit mixed sheaf construction and of the cardinality formulas? The repository's two-valued nongluing result is choice-free, whereas the present report deliberately works in ZFC. Maximal antichains, independent ultrafilter choices, and set-sized witness choices are visible places where its proofs use choice. The diagonal existence of a pseudointersection for an explicitly enumerated base is much less demanding. These differences should be preserved rather than obscured by a single global foundational slogan. (Parallel to Q8. The earlier report's own question, whether the infinite-quotient direction of its Theorem 8.5 needs dependent choice, is not addressed by either source.)
- R9. Other ideals and index sets** (source 10, 14.9). Replace  $\text{Fin}$  by an ideal  $\mathcal{I}$  on an index set  $J$ . For the quotient  $\mathcal{P}(J)/\mathcal{I}$ , identify hypotheses under which bounded-character filter sites admit analogues of Theorem 4.2, Corollary 4.5 and Theorem 7.1. What replaces the countable diagonal realization argument when  $J$  is uncountable? The criterion proof suggests an order-theoretic version using nonzero lower bounds in the quotient algebra, which source 08's Proposition 16.2 supplies for the forward direction. The reverse comparison barrier also needs a density theorem for filters with no such lower bound; that additional requirement should not be omitted. (Parallel to Q7.)
- R10. A formal proof development** (source 10, 14.10). Formalize the complete comparison barrier and the atom classification, then the mixed-model realization theorem, with all imported results listed explicitly. Can the construction serve as a reusable interface between filter semantics and complete Boolean-valued structures? A practical order of development is: finite-modification invariance and positive adjunction; regular cones and the old-event embedding; the general completion criterion; the density of filters without pseudointersections; atoms; and finally partition mixing and the coordinate diagonal. The two-sided barrier has a relatively small mathematical dependency graph and is a better

initial formalization target than the entire sheaf-theoretic presentation. No Lean code implementing these steps exists in this report or in either source package. (Parallel to Q10.)

## 20 Conclusion

Restricting the filters is not a harmless implementation choice. For countably generated filters, the associated binary sheaf has an atomless regular-open Boolean algebra, and associated field models have global rings that are not full products of fields. For all filters, the Boolean algebra is the power set of the ultrafilter set and the global sections are products of ultrapowers.

The hierarchy between these endpoints has two exact initial thresholds. The pseudointersection number controls when the original Boolean events stop being order dense, and even when a complete event-preserving comparison becomes impossible, in either direction. The distributivity number measures the first failure of simultaneous local binary decisions in the old completion and, at the successor stages above it, separates the completions abstractly. The ultrafilter number controls when atoms and geometric points first appear. Full atomicity is governed by a separate generic-extension property. A published model shows that atomicity can arrive before the full site, and that a later atomic stage can still be a different complete Boolean algebra.

The mixing calculation ties these order-theoretic facts to first-order model theory without confusing internal truth with ordinary truth of global sections. Transfer and, at the successor stages, countable saturation of specializations survive the change of truth-value algebra, while the sharp infinitary counterexample identifies exactly what is lost at the first threshold, and the exact sizes  $\mathfrak{c}$ ,  $2^{\mathfrak{c}}$ ,  $2^{2^{\mathfrak{c}}}$  measure what mixing adds.

The report answers the bounded-character comparison, point-existence, local binary-reading and countable specialization questions formulated here. Bibliographical priority, intermediate algebra classification, higher saturation, the small-character ultrafilter extension property in ZFC terms, and the singular stages remain unestablished. The earlier repository nongluing result is not claimed as new. These are explicit conclusions about the repository's filter-site construction; they do not assert a resolution of an unrelated major set-theoretic conjecture or certify publication priority.

## References

- [1] Vladimir Reshetnikov, *ProveIt*, repository snapshot e21766d04c2b8a9b2cdba0cd43e563024b3bd1b9, inspected 24 September 2026 by both sources. Relevant unrefereed manuscript: *A two-valued obstruction to gluing Fréchet powers*, 20 September 2026; especially its countably generated-site discussion (Section 7), its full-site sheafification theorem (Theorem 9.2) and its non-claim about other filter sites (Section 10). In the current tree it is the neighbouring report `SetTheory/Cardinals/docs/reports/ordinals-and-order-types/frechet-two-valued-obstruction/`. <https://github.com/VladimirReshetnikov/ProveIt/tree/e21766d04c2b8a9b2cdba0cd43e563024b3bd1b9/SetTheory/Cardinals/docs/reports/ordinals-and-order-types/frechet-two-valued-obstruction>
- [2] Guillaume Massas, *A Semi-Constructive Approach to the Hyperreal Line*, Australasian Journal of Logic **20**(3) (2023), 490–536. DOI: 10.26686/ajl.v20i3.8254. The motivating question occurs after Fact 5.10, p. 517 (source 08 inspected the page image). The filter-site definitions, sheaf discussion, and canonical-extension comparison are in Sections 5.2 and 6.1 (source 10 inspected preprint pages 23, 25 and 27 as page images). Author's preprint: arXiv:2201.10818 (2022). <https://ojs.victoria.ac.nz/ajl/article/download/8254/7451>; <https://arxiv.org/abs/2201.10818>

- [3] Andreas Blass, *Combinatorial Cardinal Characteristics of the Continuum*, in M. Foreman and A. Kanamori (eds.), *Handbook of Set Theory*, Springer, 2010, 395–489. Author-hosted draft dated 24 November 2003 was consulted for background by both sources; it is not represented as the final published text. Used for standard terminology and historical context, not as a source of a claimed new inequality. All properties of  $\mathfrak{p}$ ,  $\mathfrak{h}$  and  $\mathfrak{u}$  needed in the proofs are defined and proved in the text. <https://sites.lsa.umich.edu/ablass/wp-content/uploads/sites/1471/2025/10/hbk.pdf>
- [4] Moreno Pierobon and Matteo Viale, *Boolean valued models, presheaves, and étalé spaces*, arXiv:2006.14852, version 6, 23 July 2026. Used for the established relationship between Boolean-valued mixing and dense-topology sheaves, and for the distinction between ordinary and complete Boolean morphisms. <https://arxiv.org/abs/2006.14852v6>
- [5] Andrés Millán, *Remarks on the Generic Existence of Ultrafilters on  $\omega$* , *Topology Proceedings* **34** (2009), 27–37. Definition 2.1 gives generic existence; Theorem 3.14 and its proof, pp. 34–35, supply the model used in Proposition 9.1. <https://topology.nipissingu.ca/tp/reprints/v34/tp34003.pdf>
- [6] Douglas Ulrich, *Keisler’s Order and Full Boolean-Valued Models*, arXiv preprint 1810.05335, version 1 (2018). Sections 2–3 provide the full Boolean-valued model and specialization framework; the localized realization and specialization arguments used here are written out independently. <https://arxiv.org/abs/1810.05335>
- [7] The Stacks Project Authors, *The Stacks Project*, Sites and Sheaves, Section 7.10, “Sheafification”, tag 00W1. Accessed 24 September 2026 by source 10. <https://stacks.math.columbia.edu/tag/00W1>

## A Provenance and the merge record

### A.1 The two manuscripts

This report was built from two manuscripts of batch 36 of ProveIt’s incoming-reports intake. Their archives arrived in commit e13affd32 and were placed in commit 1a1396d4d. Both are dated 24 September 2026, both were prepared with ChatGPT, and both are pinned to ProveIt commit e21766d04c2b8a9b2cdba0cd43e563024b3bd1b9.

- **Manuscript 08** (archive `filter_complexity_research.zip`, inner directory `filter_complexity_article/`, main file `article.tex`, 24-page PDF), *Filter Complexity and Booleanization*. It is the base. It contributed the  $\mathbb{P}_\kappa$  indexing with its finite-generation and limit stages, the concrete representation  $\mathbb{C}_\kappa = \text{RO}(\omega^*, \tau_\kappa)$  and the sheaf equivalence, the forward threshold with its meet obstruction, the arity invariant and the interior formula, the atom, point and atomicity results, the closure for regular  $\kappa$ , the imported-model three-stage hierarchy, the finite-cofinite toy model, the mixing sheafification at every stage with its product formula at every atomic stage, transfer, the  $\mathfrak{p}$ -fold infinitary separation, the stable-range transport, the global rings, the general Boolean-algebra threshold and stage-comparison criterion, the finite verification suite, and questions Q1–Q10.
- **Manuscript 10** (archive `Frechet_Cardinal_Boundaries.zip`, inner directory `Frechet_Cardinal_Boundaries/`, main file `Frechet_Cardinal_Boundaries.tex`, 26-page PDF), *Three Cardinal Boundaries for Fréchet-Power Completions*. It contributed the poset-side description, the general completion criterion for admissible sites, the reverse barrier and hence the two-sided statement, the closure at successor stages with partition distributivity and local decisions, the  $\mathfrak{h}$  boundary and the unlabeled non-isomorphism, the residual formula, the no-map-into-a-power-set corollary, the forcing

reading, the infinite-labels example and simultaneous raw representatives, ordinary specializations, localized countable realization and countable saturation, the exact sizes, the consolidated picture with its CH and separated regimes, and questions R1–R10. It supplies no computational check.

## A.2 Choices made by the merge

- *Base and indexing.* Source 08 is the base because its index family strictly contains source 10's stages and its threshold, product and general-algebra results are stated more generally. Source 10's indexing was translated by Lemma 2.2 ( $\mathcal{F}_\lambda = \mathbb{P}_{\lambda^+}$ ), which the merge proves.
- *Closure-based results.* Source 10's results that depend on its closure theorem are stated at the successor stages only, with the coverage of  $\mathbb{P}_\omega$  and the non-coverage of limit stages recorded in Section 5.1. The reverse barrier uses no closure and is stated at every  $\kappa > \mathfrak{p}$ , with the reason given after Theorem 4.4.
- *One-sided versus two-sided.* Source 08's barrier is one-sided and is printed as such (Theorem 4.3); the reverse direction and the two-sided Corollary 4.5 are source 10's.
- *Shared results printed once.* Where both sources prove a result the printed proof is source 08's, except that Theorem 4.2, the failure of (v) through the dense join  $\mathcal{E}_F$ , and the poset-side atom proof are kept as source 10's genuinely different routes. Source 10's sheafification proof (Remark 11.5), its point lemma, its full-site product and its independent family repeat source 08's arguments and are credited without a second copy.
- *Merge-written text.* Lemmas 2.2 and 3.6, Sections 1.2, 1.5 and 1.6, the merge notes, the closure-scope discussion and this appendix are new. Every other statement and proof is taken from a source, translated where needed.
- *Notation.* Source 08's symbols are kept, except that its comparison map  $h$  is renamed  $J$  (to keep  $\mathfrak{h}$  free), its field  $K$  is renamed  $\mathbb{F}$  (to keep  $K_F$  free), and its abbreviation  $\mathbb{C} = \mathbb{C}_\kappa$  in the mixing section is written out (a bare  $\mathbb{C}$  is source 10's  $\mathbb{C}_\omega$ ). Source 10's renamed symbols are listed in Section 1.6. No normalization changed.
- *Labels.* Every label carries the prefix `fbc:`. Source 08's labels keep their keys after the prefix, except `sec:model`, which is `fbc:sec:early-atomicity` (source 10 used the same key for a different section), and `thm:sheafification`, which is `fbc:thm:mix-sheafification` (the earlier report uses that key for its full-site theorem).
- *Questions.* Source 10's research questions 14.1–14.10 are R1–R10. Source 08's Q4 and Q6 and source 10's R2 are re-scoped where the other source answers them.

## A.3 Checks made at intake and in the write

At intake the imported properties of Proposition 9.1 were compared with Millán's Definition 2.1 and the proof of his Theorem 3.14; they are stated there as source 08 reports them (Millán himself cites an earlier paper for them). The Pierobon–Viale version 6 of 23 July 2026 exists on arXiv. Both manuscripts were read in full at intake, including the admissible criterion, both barrier directions, the atom and point proofs, the saturation proof and the size counts, and no mathematical error was found. Neither contradicts the other: every shared statement agrees after the index shift. In the write, the proofs of the reverse barrier and of the residual formula were re-read for hidden uses of closure, and the index shift was proved.

Both sources state that they made no change to the repository; source 08 also states that nothing was submitted for publication and no source author was contacted, and that its repository audit did not cover

other repository projects theorem by theorem. Source 10 redistributes no third-party PDF or font files. Both sources compared themselves with the earlier report at the pin; that report's text was unchanged from the pin to the placement commit.

#### A.4 Concordance of numbers

| Source number (delivered PDF)           | This report                                |
|-----------------------------------------|--------------------------------------------|
| 08 Def. 2.1; 10 §2.1                    | Definition 2.1                             |
| 08 Def. 3.1                             | Definition 3.1                             |
| 10 Remark 2.3                           | Remark 3.5                                 |
| 08 Lemma 2.2; 10 eq. (1)                | Lemma 2.3                                  |
| 08 Lemma 2.3; 10 Lemma 2.1              | Lemma 2.4                                  |
| 08 Lemma 3.2                            | Lemma 3.2                                  |
| 08 Thm 3.3                              | Theorem 3.3                                |
| 10 Lemma 2.2                            | Lemma 3.4                                  |
| 08 Lemma 4.1; 10 Lemmas 4.2, 7.2        | Lemma 4.1                                  |
| 10 Thm 3.2                              | Theorem 4.2                                |
| 08 Thm 4.2; 10 Thm 4.3(a),(b)           | Theorem 4.3                                |
| 10 Thm 4.3(c)                           | Theorem 4.4                                |
| 10 Thm 4.3 (whole)                      | Corollary 4.5                              |
| 08 Cor. 4.3; 10 eq. (9)                 | Corollary 4.6                              |
| 10 Example 4.4                          | Example 4.7                                |
| 08 Def. 4.4, Prop. 4.5, 4.6             | Definition 4.8, Propositions 4.9, 4.10     |
| 08 Prop. 5.5                            | Proposition 5.1                            |
| 10 Thm 5.1, Cor. 5.2                    | Theorem 5.2, Corollary 5.3                 |
| 10 Def. 6.1, Thm 6.2, Cor. 6.3          | Definition 6.1, Theorem 6.2, Corollary 6.3 |
| 08 Thm 5.1; 10 Thm 7.3, Cor. 7.4        | Theorem 7.1, Corollary 7.6                 |
| 08 Prop. 5.2; 10 Lemma 7.5, Cor. 7.6    | Proposition 7.2, Corollary 7.3             |
| 08 Thm 5.3; 10 Thm 7.3                  | Theorem 7.5                                |
| 08 Remark 5.4; 10 Remark 7.7            | Remarks 7.7, 7.4                           |
| 10 Prop. 8.1                            | Proposition 8.1                            |
| 08 Prop. 6.1, Lemma 6.2; 10 Lemma 12.3  | Proposition 9.1, Lemma 9.2                 |
| 08 Thm 6.3                              | Theorem 9.3                                |
| 08 Prop. 7.1                            | Proposition 10.1                           |
| 08 Prop. 8.1                            | Proposition 11.1                           |
| 08 Def. 8.2                             | Definition 11.2                            |
| 08 Lemmas 8.3, 8.4; 10 Lemma 9.1        | Lemmas 11.3, 11.4                          |
| 08 Thm 8.5; 10 Thm 9.2                  | Theorem 11.6                               |
| 08 Cor. 8.6; 10 Prop. 9.3               | Corollary 11.7                             |
| 10 Example 9.4, Prop. 9.5               | Example 11.8, Proposition 11.9             |
| 08 Cor. 8.7; 10 Prop. 12.1              | Corollary 11.10                            |
| 08 Thm 9.1; 10 Thm 10.1                 | Theorem 12.1                               |
| 10 Remark 10.2, Prop. 10.3              | Remark 12.2, Proposition 12.3              |
| 08 Thm 9.2, Cor. 9.3                    | Theorem 12.4, Corollary 12.5               |
| 10 Thm 11.1, 11.2                       | Theorems 13.1, 13.2                        |
| 08 Thm 10.1, Cor. 10.2; 10 Example 11.3 | Theorem 14.1, Corollary 14.2               |
| 10 Lemma 12.2, Thm 12.4                 | Lemma 15.1, Theorem 15.2                   |
| 08 Def. 11.1, Prop. 11.2, 11.3          | Definition 16.1, Propositions 16.2, 16.3   |
| 08 §12; 10 §15                          | Section 18                                 |
| 08 Q1–Q10; 10 Questions 14.1–14.10      | Q1–Q10; R1–R10 (Section 19)                |